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RESEARCH ARTICLE

XAI-SkinCADx: A Six-Stage Explainable Deep Ensemble Framework for Skin Cancer Diagnosis and Risk-Based Clinical Recommendations

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ABSTRACT The fastest growing and deadliest cancer, skin cancer, requires diagnostic solutions that are accurate, understandable, and therapeutically actionable. Our six-stage deep ensemble diagnostic framework, XAI-SkinCADx, employs hybrid feature extraction, deep learning ensembles, understandable AI, and clinical recommendation creation to meet this need. The 2367 dermoscopy images belonging to nine distinct classes of the ISIC skin cancer dataset are used to train the framework. Adding 5,023 new images makes the dataset richer and better balanced across the classes. A comprehensive performance evaluation is ensured by using the dataset with an 80:20 train-data/test-data split. It begins with hand-designed feature extraction using GLCM-based dimensionality reduction and visualization and Local Binary Patterns (LBP) visualization. Three convolutional neural networks are then employed to extract specific spatial features: DV-25 (DenseNet201 + VGG16), DE-25 (DenseNet201 + EfficientNetB5), and DM-25 (DenseNet201 + MobileNetV2). Temporal patterns are described with bidirectional long-short-term memory (BiLSTM) and finally classified with multiclass SVM. The accuracy is 95.63%, precision is 96.2%, recall is 95.7%, F1 score is 95.9%, and area under curve is 0.97. We interpret it using Grad-CAM++ and LIME. Compared to Grad-CAM++, LIME had better localization accuracy (IoU: 0.78, Dice: 0.87) while Grad-CAM++ had worse accuracy (IoU: 0.70, Dice: 0.81). The best performing classes with LIME interpretability, Vascular Lesion, Seborrhoeic Keratosis, and Dermatofibroma, respectively, with Dice scores of 0.91, 0.89, and 0.87 respectively. For providing patients with personalized suggestions independent of other information, a dermatology-focused LIME-based recommendation platform was designed. The timely dermatology consultation and decreased clinical burden are led by the classification by the system of the disorders as low-risk, medium-risk, and high-risk. By integrating accuracy, clarity, and utility, XAI-SkinCADx creates a new benchmark for applying AI in the diagnosis of skin disease, enabling physicians to make well-informed and trusted decisions.

INDEX TERMS Skin cancer diagnosis, hybrid CNN, grad-CAM++ visualization, LIME interpretability, deep ensemble learning, explainable artificial intelligence (XAI), recommender system.

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I. INTRODUCTION

Skin cancer is the fastest-growing disease today. It is a big public health concern all around the globe because it is

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increasingly becoming common, is of several forms, and can develop rapidly if left untreated. Based on the World Health Organization (WHO) and Global Cancer Observatory (GLOBOCAN 2022), it is estimated that there are over 120,000 skin cancer-related deaths yearly, and over 1.5 million new diagnoses of non- melanoma and melanoma skin cancer each year [1], [2]. The most lethal form of skin cancer is melanoma, despite being less frequent compared to other forms. This is because it can spread to other regions of the body and is aggressive [3]. The incidence of skin cancer is the worst in regions with high exposure to ultraviolet radiation, such as the US, Australia, New Zealand, and part of Europe [4]. Such an increase is due to the fact that fair-skinned populations are more susceptible to UV-induced DNA damage [5]. Skin cancer incidence has been much higher in the equatorial and high-altitude areas, mainly because of environmental causes such as excessive exposure to the sun, ozone depletion, and tanning for leisure purposes [6]. Genetic predisposition, compromised immunity, and faulty practices of skin protection, in addition to occupational exposure to toxins including arsenic and industrial waste, further speed up this rising trend [7]. Lack of proper public awareness and the non-availability of preliminary screening centers in the tropics and subtropics further aggravates the mortality rates resulting from late diagnosis [8]. Spread of diseases is further dominated by socioeconomic inequalities, climate change, and aging populations [9]. This is worrying considering the American Cancer Society projects an even annual increase in the number of melanomas, to the tune of 100,640 new diagnoses and well over 8,000 melanoma-related deaths alone in the United States in the year 2024 [10]. Where dermatologic proficiency is rare in settings with meager resources, the aforementioned demographics accentuate the critical need for innovative, affordable, and interpretable diagnostic aids most specifically AI-assisted solutions that can enable early and accurate diagnosis [11].

Despite the previously held belief that skin cancer had been relatively rare in India because of the protective component of melanin-rich skin, it is now showing an alarming rise all over the country. Within the exposed sites like the face, ears, neck, and hands, the symptoms are typically seen as new lesions, changing lesions, non-healing ulcers, leathery patches, or shiny bumps [12]. As per the Indian Council of Medical Research (ICMR), the incidence of the overall non-melanoma skin cancer is lower compared to those of Western countries. Increased rates of incidence among areas like Northeast and East India are being seen, with the rates of non-melanoma skin cancer increasing to 6.2 per 100,000 among the males of the East [13]. As the ozone layer is being depleted, increasing UV exposure, changing lifestyles leading to an increase in the frequency of outdoor exposure with no use of sun protective measures, and environmental toxins like arsenic consumption through contaminated groundwater, especially in West Bengal, have been linked to the causation of skin cancer, and all of them are contributing factors to the rising incidence [14]. Apart from this, the use of 'kangri' containers among other customs has been responsible for localized forms of skin cancer, Kangri cancer [15]. The accessibility to dermatology is still limited in India, with only around 16,400 registered dermatologists, leading to the ratio of one dermatologist to 140,000 citizens. This is even lesser in rural regions despite the rising demand for the same. The immediate need for increased access to dermatology, preventive care, and escalating awareness to reduce the rising incidence of skin cancer is stressed upon [16].

Artificial intelligence has emerged as a solution to dermatology for a country like India, with the ratio of dermatologists to the population being distressingly low and specialized skin cancer care feasible only in urban areas [16]. Employing multiple trained neural network algorithms to detect numerous visual disease signs of melanoma, basal cell carcinoma, and benign ones such as acne and dermatitis, advanced AI methods, namely deep ensemble methods, provide very high-precision outputs [17]. AI platforms are now able to interpret dermoscope images with the same level of accuracy expected of dermatologist experts through computer vision and medical image analysis [18]. These platforms utilize explainable AI (XAI) methods such as Grad-CAM++ and LIME, allowing clinicians to visualize and have confidence in the model's decision through the highlighting of the relevant skin areas responsible for the predictions [19]. This transparency is important in medical applications to ensure that AI serves as a trusted decision-support system and not an enigmatic system. Additionally, dermatology-specific recommender platforms can provide personalized treatment pathways and triage advice based on the level of hurry, thus allowing for the smooth triaging of patients from towns and cities to further treatment centers, especially those with limited resources, by healthcare experts [20]. The platforms are capable of identifying the initial signs of skin pathology such as acne, lesions, and alterations in pigments and classifying them accordingly under risk levels for further evaluation. AI-assisted diagnostic platforms have the promise to decrease waiting time for patients drastically, ensure regular routine screening programs, and bridge the urban-rural healthcare divide. On the front of overall medical specialties, AI has shown promising success already through oncology, pathology, and radiology. Its application is expected to speed up diagnosis, minimize the role of errors committed by humans, and provide scale-up and affordable skincare, especially among nations where there is not enough specialty talent [21].

The study discusses on a six-stage comprehensive framework that may help doctors correctly identify skin disorders and provide each patient personalised therapy recommendations based on their specific ailment. At the beginning, special methods like Local Binary Patterns (LBP) and Grey-Level Co-occurrence Matrix (GLCM) are used to get the hybrid features. These methods assist show the texture of the skin lesions in high resolution. This is followed by the

next stage where features are being collected through integrated Convolutional Neural Network (CNN) combinations such as the combination of DenseNet201 with VGG16, EfficientNetB5, and MobileNetV2 to enhance detection accuracy across lesion types. The model can incorporate context and sequential patterns with the addition of temporal learning using Bidirectional Long Short-Term Memory (BiLSTM) in the third step. The classifying is finally completed through the use of multiclass support vector machine (SVM) to deliver high-performance results including precision, accuracy, and AUC. The fifth stage incorporates explainable AI methods such as the use of Grad-CAM++ and LIME to visualize and understand the areas of interest affecting model predictions and provide transparency and clinician trust. The sixth step uses a dermatology-specific recommender system to analyze the output and provide clinical action plans and disease-specific urgency ratings. The end-to-end system enhances diagnostic accuracy and addresses the practical need for interpretable and sharable AI in dermatology treatment.

II. LITERATURE SURVEY

Skin cancer detection has been significantly enhanced by the implementation of recent AI innovations, such as deep learning, ensemble methodology, and explainable AI (XAI) techniques. These developments seek to enhance diagnostic accuracy, interpretability, and clinical decision support. The Table 1 outlines important 2022 to 2025 research, noting methodologies, datasets, evaluation measures, strengths, limitations, and areas of future research.

Some more investigations, for example, SkinDWNet [31], SNC-Net [32], DVFNet [33], SCDNet [34], Xception–ResNet101 concatenation [35], federated and transfer learning techniques [36], and DSCC-Net [37], have further improved automatic diagnosis of skin cancer, along with the investigations listed within Table 1. By consistently demonstrating accuracies of 92–96% or more upon benchmarking corpora such as ISIC, HAM10000, and Derm7pt, such techniques identify the continuing necessity for hybrid combination of features, ensemble architecture, and federated methodologies. Nevertheless, as the comparative findings within Table 1 exemplify, such models remain accuracy-led throughout with little or no integration of interpretable AI mechanisms and little practical recourse for clinical recommendation or decision support. They further question the generalisability and robustness of their findings amongst several populations within dermatological practice because they depend upon single corpora alone. Compared with this, the XAI-SkinCADx framework we outline not only achieves and exceeds many such benchmarks (accuracy = 95.63%, F1 = 95%, AUC = 0.97) but also embeds within it a double-layer interpretability framework (Grad-CAM++ and LIME) together with a clinically oriented recommender system. This concurrent emphasis upon enablist direction and explainability advances our work as an entire framework filling the vacuities identified within prior investigations and therefore better aligning it with the requirements of practical dermatological work within the real world.

A. RESEARCH GAP

In spite of the notable developments in AI-powered dermatologic diagnostic techniques between 2022 and 2025, existing literature identifies several key gaps for further research to enable clinical integration. Bibi et al. [22] made high accuracy using an ensemble of conventional classifiers and genetic optimization but is computationally demanding and not ready for real-time deployment. The SkinWiseNet team [23] reported outstanding results on varied skin types, but there is the risk of overfitting, and fewer population-scale verifications have been performed. The integration complexity of the model presented by Akter et al. [24]. makes the system not vitally geared toward rapid, scale-up diagnostics. Liu et al. [25] have discovered that stacking MobileNetV2, ResNet18, and VGG11 only achieved 86.7% accuracy, thus more research has to be conducted on the structure. Since it is only able to conduct binary classification, it is only applicable for the diagnosis of limited skin diseases. This is true even if Shinde et al. [26] have created a cheap, IoT-compatible model with high diagnostics. Himel et al. [27] proposed Vision Transformers with segmentation but are too computer-intensive to utilize in low-resource environments. Gomathi et al. [28] proposed the utilization of dual-optimized deep network but had decreased specificity to differentiate between cancer and benign cases. Chakravorty et al. [29] detected asymmetry with the use of feature extraction but did not provide conventional metrics such as accuracy or AUC, compromising the strength of the comparison. Last but not least, Renith and Senthilselvi [30] suggested an optimized AdaBoost-AlexNet pipeline but it's complicated optimization hampers the ease of use. All of the aforementioned work lacks consistency with an integrated system having high diagnostic accuracy, explainability via the use of XAI, clinician interpretability, and personalized recommendation toward treatment – all integrated inside a lightweight, module-like framework. This gap attests to the need for an uncompromisingly robust, interpretable, and clinically applicable system such as XAI-SkinCADx, where ensemble learning, hybrid CNN-BiLSTM-SVM modeling, explainability using AI, and integration of dermatologic recommender are used to enable practical use inside real-life medical environments.

B. TECHNOLOGIES UTILIZED IN XAI-SKINCADX SYSTEM

To fill the gaps observed in the existing research, the proposed XAI-SkinCADx system unites the various cutting-edge technologies into an efficient six-step framework optimized for high accuracy, readability, and clinical usability. The hybrid feature extraction method of (LBP + GLCM) is used to compute strong multi-texton features, followed by Deep Ensemble CNNs (DenseNet201 is blended with VGG16, EfficientNetB5, and MobileNetV2) for spatial representation. To extract temporal dependence and improve the learning of features, a BiLSTM network is utilized, and

TABLE 1. Literature survey on Ai-Based skin cancer diagnosis and treatment planning.

Author(s)	Year	Journal	Methodology	Dataset	Accuracy	Advantages	Limitations	Research Gap
Bibi et al [22].	2025	Nature Scientific Reports	Max Voting ensemble combining Random Forest, SVM, and MLPN with Genetic Algorithm-based feature selection	HAM10000, ISIC 2018	94.70%	Enhanced predictive performance through ensemble learning	Computational complexity due to multiple models	Need for real-time deployment in clinical settings
SkinWiseNet Team [23]	2024	BMC Medical Informatics and Decision Making	Feature fusion from diverse datasets to reduce bias in skin tone and hair presence	Multiple skin lesion datasets	99.86%	High accuracy across diverse skin tones and conditions	Potential overfitting to training data	Validation on larger, varied populations needed
Akter et al [24].	2024	arXiv	Hybrid Deep Learning model combining InceptionV3 and DenseNet121 with weighted sum rule for feature fusion	ISIC dataset	92.27%	Robust classification with high sensitivity and specificity	Complexity in model integration	Simplification for broader adoption
Liu et al [25].	2024	arXiv	Ensemble learning using stacking of MobileNetV2, ResNet18, and VGG11	HAM10000	86.70%	Improved accuracy over individual models	Moderate accuracy compared to state-of-the-art	Exploration of more advanced ensemble techniques
Shinde et al [26]	2023	Research Gate	Squeeze-MNet: Lightweight model using digital hair removal and MobileNet for IoT devices	ISIC dataset	99.36%	High accuracy with low computational requirements	Limited to binary classification	Expansion to multiclass scenarios
Himel et al [27]	2024	arXiv	Vision Transformer with Segment Anything Model for segmentation and classification	HAM10000	96.15%	Superior performance with self-attention mechanisms	High computational resources required	Optimization for real-time applications

TABLE 1. *(Continued.)* **Literature survey on Ai-Based skin cancer diagnosis and treatment planning.**

Gomathi et al [28]**.**	2023	Biomed Signal Processing	Dual optimization-based deep learning network for skin cancer detection	PH^2 dataset	85.70%	Effective early detection system	Lower specificity (60%)	Enhancement of specificity metrics
Chakravorty et al [29]**.**	2023	Journal of Investigative Dermatology	Asymmetry classification using dermatologist-like feature extraction	PH^2 dataset	Not specified	Improved asymmetry classification	Lack of comprehensive performance metrics	Inclusion of accuracy and AUC values
Renith & Senthilselvi [30]	2023	PMC	Improved AdaBoost with Aphid-Ant Mutualism optimization on AlexNet architecture	HAM10000	95.70%	High precision and F-measure	Complexity in optimization algorithms	Simplification for clinical implementation

final classification is improved using a Multiclass Support Vector Machine (SVM). For explainability, Explainable AI techniques Grad-CAM++ and LIME—emphasize decision-relevant skin areas, bridging the opaque model nature of AI. Last but not least, dermatology-specific recommender system is integrated to convert model outputs to actionable clinical advice. All of the technologies together provide support for scalability, diagnostic transparency, and personalized guidance facing the drawback of earlier models head-on across performance, explainability, and deployability.

III. METHODOLOGY

The solution involves XAI-SkinCADx, a multi-stage computer-aided diagnosis (CADx) system that is comprehensible and is designed to enhance the precision, transparency, and clinical utility of skin cancer diagnosis. Figure 1 indicates that XAI-SkinCADx consists of a chain of six steps working together to predict skin lesion classifications and recommend treatments from beginning to end. The system begins using hybrid handcrafted feature extraction for texture analysis. It follows this by utilizing a collection of integrated convolutional neural networks (CNNs) for deep spatial representation., a Bidirectional Long Short-Term Memory (BiLSTM) network digests these features to discover patterns in time and in context. A multiclass support vector machine (SVM) is used in the final classification in order to make the decision boundaries strong for every class of lesion. In order to satisfy the most critical requirement in the healthcare sector, that is, explainability, the system employs Explainable AI (XAI) techniques such as Grad-CAM++ and LIME to visualize the regions on the skin most critical for the prediction by the model [38], [39]. A recommendation module with a dermatology background finally converts a diagnosis output into actionable clinical advice depending on the kind of disorder and how severe it is. XAI-SkinCADx is an end-to-end diagnostic tool for applied dermatology applications since it merges deep learning, transparency, and clinical expertise in a meaningful manner [40].

A. DATASET

We employed the publicly accessible ISIC Skin Lesion Dataset to develop and test the suggested XAI-SkinCADx diagnostic system. The skin images in the dataset are categorized into nine groups: actinic keratosis, basal cell carcinoma, dermatofibroma, melanoma, nevus, pigmented benign keratosis, seborrhoeic keratosis, squamous cell carcinoma, and vascular lesion [41]. There are a total of 2,367 images in the dataset. They are allocated as follows: 2,249 images for training and 118 for testing. We apply an 80:20 split method for ensuring equitable validation of the model. Figure 2 illustrates the class distribution in both the training and test subsets. It is evident from the figure that there is an imbalance among the classes, particularly in classes such as seborrhoeic keratosis and vascular lesions [42]. Figure 3 depicts sample images of every class of diseases in the training set. The figure illustrates how lesions may present differently in different categories. Table 2 provides a complete identification of the number of images by class, highlighting both training and testing distributions. We do this for clean and reproducible results. The distribution here indicates we have a need for effective methods to extract features and learn that will work

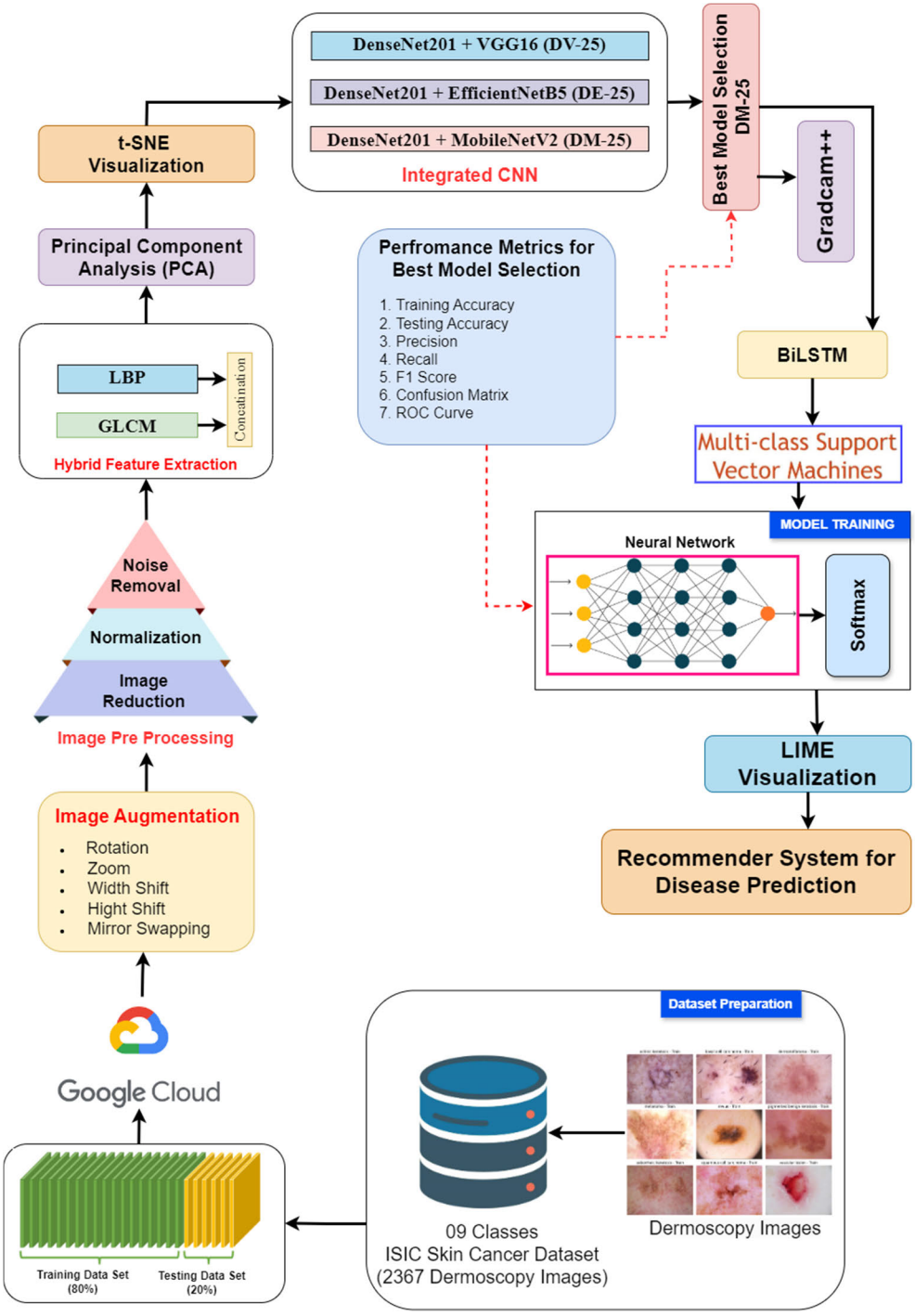


FIGURE 1. The Architectural pipeline of XAI-SkinCADx.

well for not-well-represented classes. The large number of images with clinical notes provides a robust basis for training the proposed ensemble deep learning models, which are able to detect all skin cancer lesion types, whether common or rare. We applied, prior to model training, data preprocessing, augmentation, and class-balancing techniques in an attempt to address the differences that were present in the dataset.

B. DATASET UTILIZED IN GOOGLE CLOUD

The ISIC Skin Cancer dataset was loaded onto Google Drive and directly accessible through Google Colab Pro+, facilitating a cloud-based pipeline for training the envisaged XAI-SkinCADx framework. After division based on a 80:20 train:test ratio, the dataset was augmented with rotation, zoom, shift, and flip operations to achieve greater diversity for

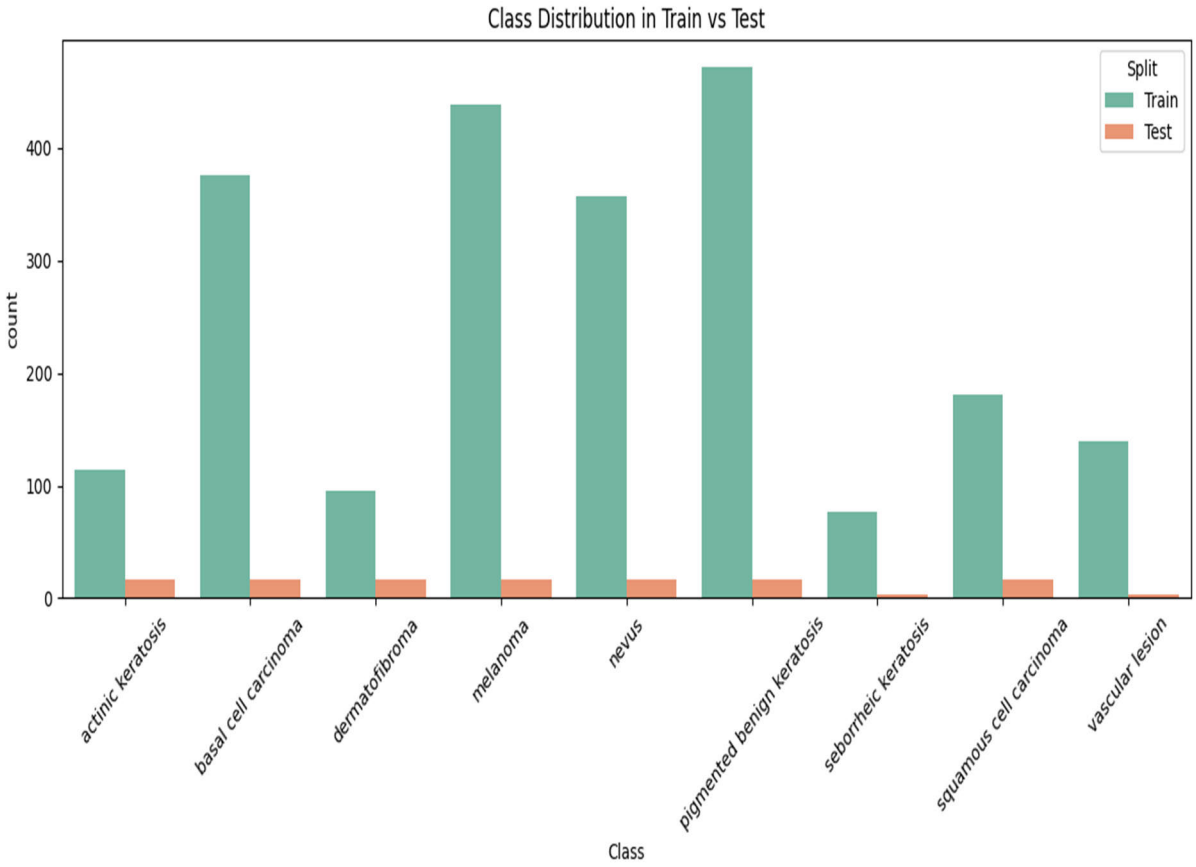


FIGURE 2. Distribution of datasets in different classes, including train and test.

TABLE 2. ISIC Dataset Class Distribution.

Class	Train Images	Test Images	Total Images
Actinic Keratosis	114	16	130
Basal Cell Carcinoma	376	16	392
Dermatofibroma	95	16	111
Melanoma	438	16	454
Nevus	357	16	373
Pigmented Benign Keratosis	472	16	488
Seborrheic Keratosis	77	3	80
Squamous Cell Carcinoma	181	16	197
Vascular Lesion	139	3	142
Total	**2249**	**118**	**2367**

training. All the images were re-sized to 224 × 224 pixels and normalized to allow for the usage of pre-trained CNNs [43]. Integrated CNN feature extractions were sent to XGBoost for further refinement by Multiclass SVM. The framework also utilized LIME for visual interpretability along with a clinical recommender module, run within a reproducible, scalable cloud setting [44].

C. AUGMENTATION AND PRE-PROCESSING

In an effort to enhance the model's generalisability and robustness, a pre-processed and augmented dataset for skin cancer from the ISIC dataset was built. These steps yielded a total of 5,023 images, distributed as follows: 4,024 for training and 999 for testing, maintaining a consistent train-test split of 80:20. The class distribution within both the training and test subsets was balanced effectively, as revealed in Figure 4. All nine of the diagnosis categories actinic keratosis, basal cell carcinoma, dermatofibroma, melanoma, nevus, pigmented benign keratosis, seborrhoeic keratosis, squamous cell carcinoma, and vascular lesion—had an identical number of images: 444 for training and 111 for testing. This is a consideration in ensuring consistency of the model performance, particularly for the multi-class problem, as it prevents class imbalance [45]. Table 3 gives a numeric class breakdown by image numbers for every class by way of this augmentation. All images underwent an exhaustive pre-processing pipeline in preparation for training. To meet the CNN model requirements, every image was first resized into a fixed input resolution (e.g., 224 × 224 pixels). Bilateral

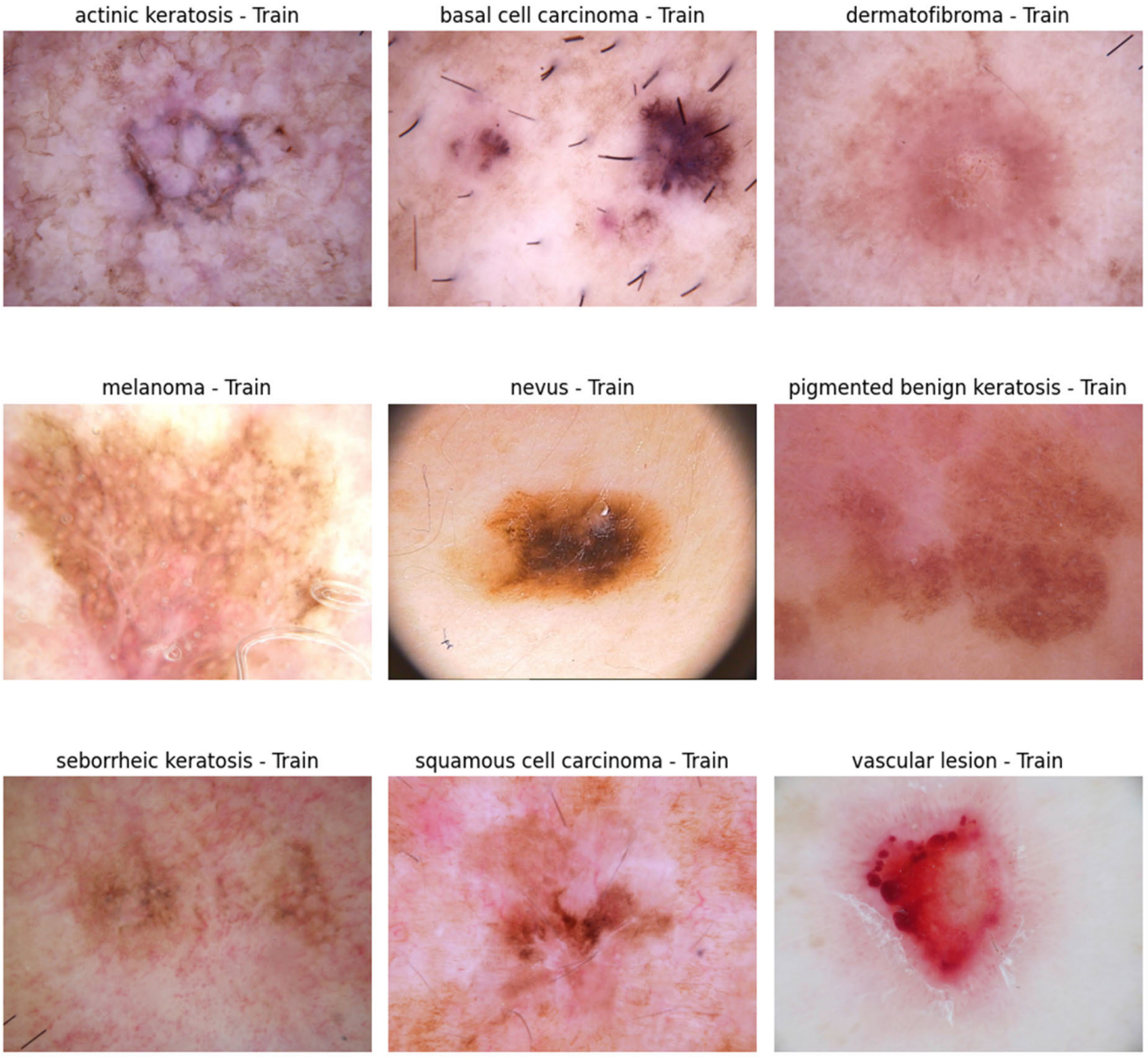


FIGURE 3. Sample training Datasets images.

filtering, with the parameters diameter = 9, sigmaColor = 75, and sigmaSpace = 75, is used to remove noise. The process preserves edges while smoothing homogeneous regions. Histogram equalisation is applied to the HSV colour space's V channel to improve lesion prominence and lesion contrast response against a change in lighting conditions [46]. An ensuing reconversion from RGB follows the step above. The training behavior was finally stabilized among cohorts by normalizing pixel intensities to [1, 0]. More sophisticated strategies for combating overfitting and enhancing the richness of the data were attempted in an endeavor to improve the utility and robustness of the dataset, including using the following: rotation_range = 20°, zoom_range = 0.2, width_shift_range = 0.15, height_shift_range = 0.15, and, in the event of a horizontal image, a transformation through the parameter indicating a mirror swapping of axis y and axis x. These adjustments considerably better allow the dataset to accommodate a range of real-world variability in skin lesion images [47]. Figure 5 reveals the enhanced uniformity, clarity, and utility in diagnosis in the dataset obtained through the enhancing pipeline, as exemplified through a set of sample images pre-processed.

A multi-stage protocol was strictly followed within this study for ensuring the proper classification of dermoscopic images. Stage I consisted of normalizing all the input through preprocessing by fixing the image size at 224 × 224 pix-

els, denosing via bilateral filtering, sharpening the edges of the lesion via histogram equalization, and normalizing the pixel intensity to [1, 0]. In a bid to tackle class imbalance and promote generalization, data augmentation techniques were employed, for example, controlled rotations (20°), enlargements (0.2), horizontal and vertical shifts (0.15), and invert. This led to a balanced dataset with nine lesion classes. The feature extraction methodology took a hybrid approach, wherein handcrafted features were developed via Local Binary Patterns (LBP) for extracting small scale differences within textures and a Grey-Level Co-occurrence Matrix (GLCM) for measuring the co-occasion of the spatial relationship of intensities via contrast, correlation, energy, and homogeneity. Concurrently, multi-scale representations of lesions were preserved through the integration of convolutional neural networks (DenseNet201 or VGG16 or EfficientNetB5 or MobileNetV2 combination) for extracting deep features. In order to encode contextual dependency over structures of lesion, these heterogenic features were stacked into an aggregate vector, reduced via Principal Component Analysis (PCA) and maintaining the first 98% of the total variance and then modelled via Bidirectional LSTM (BiLSTM). Finally, a multiclass SVM within an RBF kernel was carried out for creating stable decision borders. This methodology proved capable of discriminating classes that were visually overlapping, for e.g., melanoma and pigmented benign keratosis. This single methodology of the preprocess, feature extract, and classify ensured interpretable results for clinical purposes and high prediction accuracy.

The ISIC database used for analysis purposes had 2,367 dermoscopic images, which were spread over a total of nine classes of lesion. The images were initially separated into training (2,249) and testing (118) sets based on a ratio of 80:20. Augmentation techniques like rotation (20°), zoom (0.2), changes in width/height (0.15), and flipping were created for the purpose of dealing with class unbalance and thus contributed an additional 5,023 images. After undergoing the process of augmentation, the classes were balanced with 444 training and 111 testing data and thus an equal spread across the classes was ensured. In the process, the diversity of the data had increased and it provided a stable base for generalizing the model.

D. STAGE 1: HYBRID FEATURE EXTRACTION

The XAI-SkinCADx pipeline's initial stage employed a hybrid feature extraction strategy to extract both handcrafted and deep representations from each dermoscopic image. To facilitate the extraction of Local Binary Pattern (LBP) and Grey-Level Co-occurrence Matrix (GLCM) features, each input image was initially resized to 224 × 224 pixels and converted to greyscale. The LBP method encapsulates local texture by comparing each pixel with its neighbours, resulting in a histogram-based descriptor that captures micro-textures [48]. GLCM features were computed concurrently to extract four statistical texture properties: contrast, correlation, energy, and homogeneity, which summarize the spatial relationship of pixel intensities [49].

TABLE 3. Augmented and Pre-processed ISIC Dataset Class Distribution.

Class	Train Images	Test Images	Total Images
Actinic Keratosis	444	111	555
Basal Cell Carcinoma	444	111	555
Dermatofibroma	444	111	555
Melanoma	444	111	555
Nevus	444	111	555
Pigmented Benign Keratosis	472	111	583
Seborrheic Keratosis	444	111	555
Squamous Cell Carcinoma	444	111	555
Vascular Lesion	444	111	555
Total	**4,024**	**999**	**5,023**

In order to extract deep features, a pre-trained Convolutional Neural Network (CNN) processed the RGB image and obtained the middle layers' activations. The combined final feature vector was obtained by fusing the features from LBP, GLCM, and CNN into a single description. To facilitate subsequent model training, these features were saved as (.npy) files (X_train_hybrid.npy and X_test_hybrid.npy). Table 4 provides a snapshot of the composite feature matrix for 10 samples, delineated by LBP, GLCM, and CNN dimensions, to illustrate the composition of these features [50].

Principal Component Analysis (PCA) was implemented to preserve 98% of the variance, which led to a diminished yet information-rich feature space because of the high dimensionality of the combined features. Figure 6 illustrates the efficacy of the hybrid feature representation, as evidenced by the distinct separation of classes. The discriminative ability of the reduced PCA-features is depicted through the t-distributed stochastic neighbour embedding (t-SNE). Figure 7 also illustrates how every single feature is distributed and its significance through a heatmap representing feature strengths for 20 random samples. Both small texture information and dominant overall patterns are well-preserved through this combined step of feature extraction, resulting in robust performance in the downstream classifying steps [51].

Mathematical Fusion (LBP+GLCM+CNN)

(1) Feature block:

$$\mathrm{f}_{\mathrm{LBP}}, \mathrm{f}_{\mathrm{GLCM}}, \mathrm{f}_{\mathrm{CNN}}$$

(2 Standardize → weight → concatenate → l_2 normalize:

$$\hat{f}_k = \frac{\mathrm{f}_k - \mu_k}{\sigma_k},$$

$$\mathrm{F} = \left[\alpha_{\mathrm{LBP}}\hat{f}_{\mathrm{LBP}}; \alpha_{\mathrm{GLCM}}\hat{f}_{\mathrm{GLCM}}; \alpha_{\mathrm{CNN}}\hat{f}_{\mathrm{CNN}}\right], \tilde{F}$$

$$= \frac{\mathrm{F}}{\|\mathrm{F}\|_2}$$

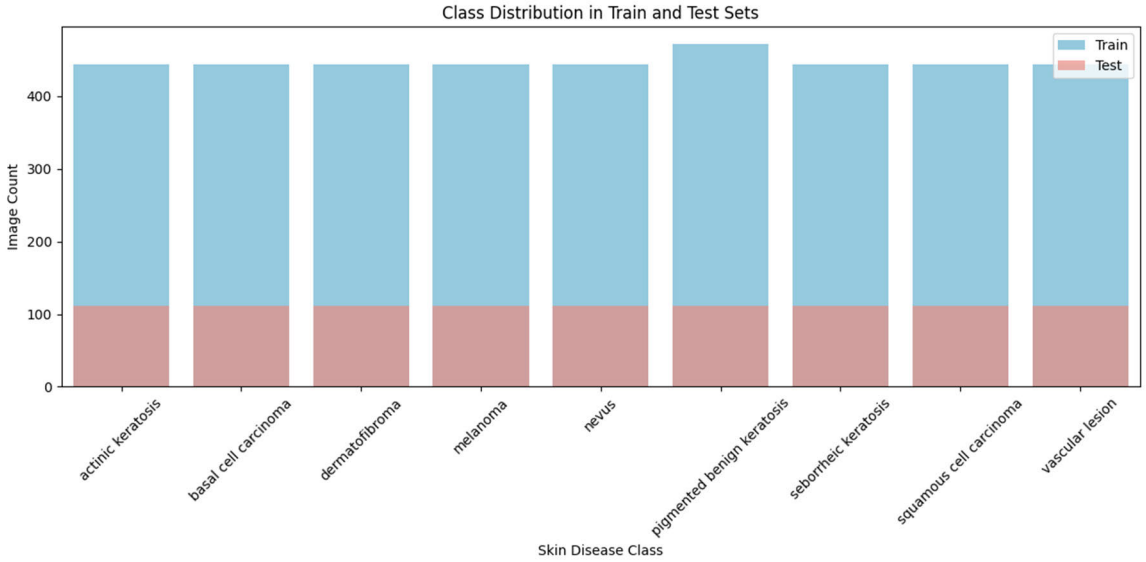


FIGURE 4. Augmented Class Distribution in Train and Test Sets.

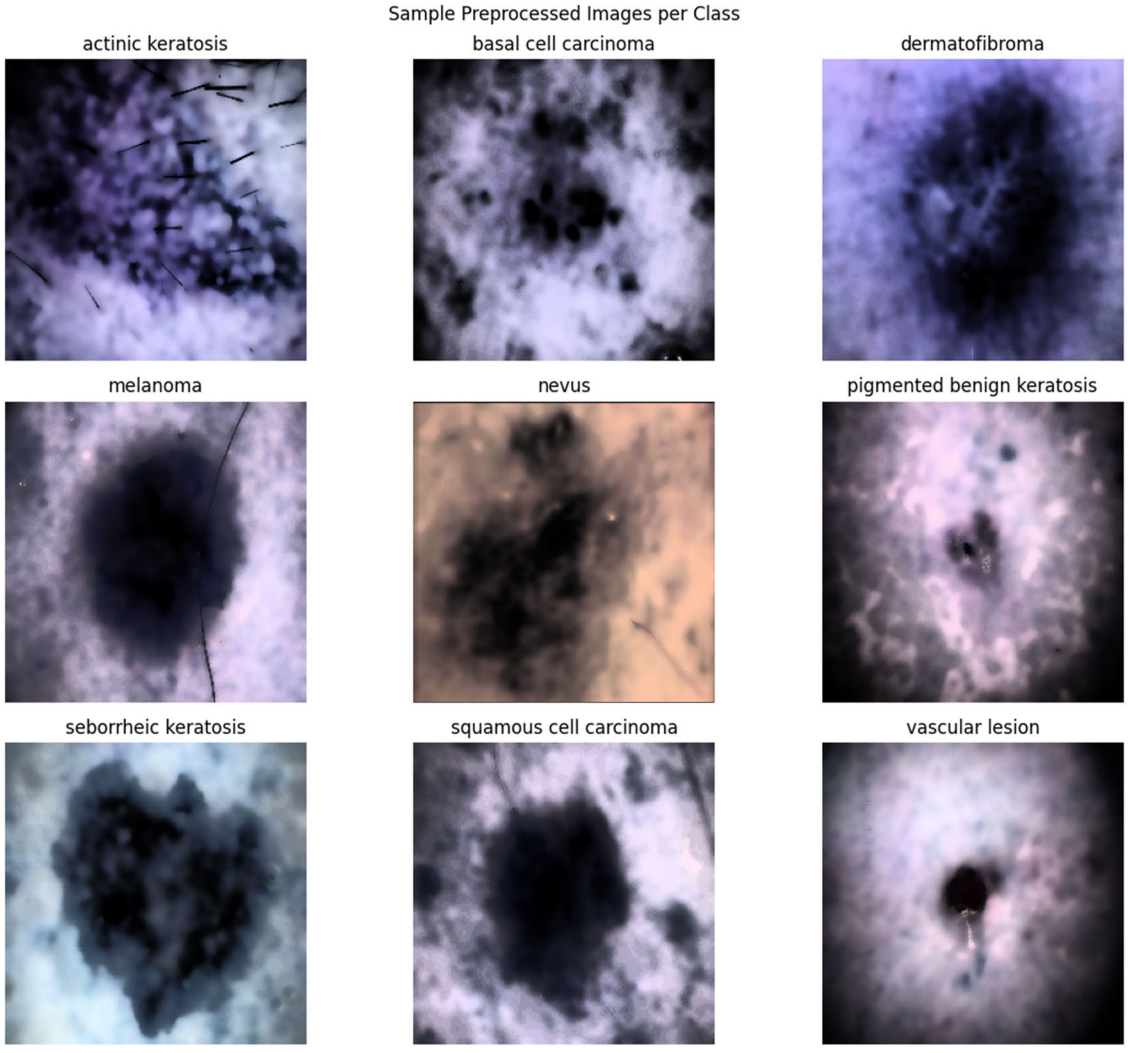


FIGURE 5. Sample Pre-processed Images per class.

TABLE 4. Feature breakdown for first 5 samples (LBP, GLCM, CNN).

LBP_0	LBP_1	LBP_2	LBP_3	GLCM_contrast	GLCM_correlation	GLCM_energy	GLCM_homogeneity	CNN_0	CNN_1	CNN_2	CNN_3	Class
0.3745	0.9507	0.732	0.5987	0.0314	0.6364	0.3144	0.5086	0.9624	0.2518	0.4972	0.3009	Vascular Lesion
0.0206	0.9699	0.8324	0.2123	0.9076	0.2493	0.4104	0.7556	0.3678	0.6323	0.6335	0.5358	Squamous Cell Carcinoma
0.6119	0.1395	0.2921	0.3664	0.2288	0.077	0.2898	0.1612	0.3411	0.1135	0.9247	0.8773	Actinic Keratosis
0.6075	0.1705	0.0651	0.9489	0.9297	0.8081	0.6334	0.8715	0.642	0.0841	0.1616	0.8986	Melanoma
0.122	0.4952	0.0344	0.9093	0.8037	0.1866	0.8926	0.5393	0.6576	0.5683	0.0937	0.3677	Vascular Lesion

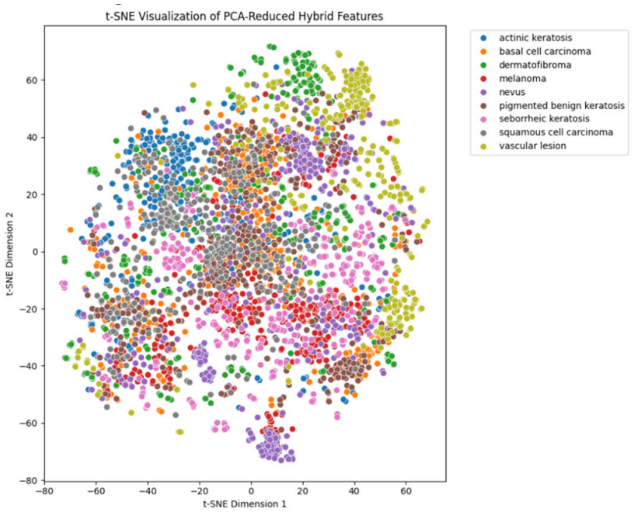


FIGURE 6. t-SNE visualization of PCA reduced features.

(3) PCA reduction (retain 98% variance):

$$z = \mathrm{W}^{\mathrm{T}}\left(\tilde{\mathrm{F}} - \mu\right)$$

E. STAGE 2: CNN INTEGRATION AND ENSEMBLE CONSTRUCTION

To enhance the generalizability and richness of feature representations for skin lesion classification, we suggest the implementation of three integrated convolutional neural networks (CNNs)—DV-25, DE-25, and DM-25—in the second stage of the XAI-SkinCADx framework. Each integrated model integrates two robust backbone architectures to capture both high-level semantic features and low-level textures. Specifically, DV-25 consists of DenseNet201 and VGG16, enabling the model to capture fine-grained lesion information while retaining spatial depth. To achieve this, DenseNet's skip-connected layers of deep layers are combined with the consistent convolutional architecture of VGG16 (Figure 8a) [52], [53]. DE-25, shown in Figure 8b, in contrast, includes DenseNet201 and EfficientNetB5, leveraging the compound scaling regimen of EfficientNet for optimal accuracy-efficiency trade-offs on a spectrum of lesion dimensions and image resolutions [54]. DM-25, the most powerful among the three, is shown in Figure 8c. It is a lean yet effective model that is able to extract salient information and ensure computational efficiency. We are able to achieve this through the combination of DenseNet201 and MobileNetV2. The outcome of this integration is the creation of better-quality feature maps, as shown in Figure 9. The diverse and sharply localized patterns of 16 representative activation channels in DM-25 suggest strong spatial attention across lesion regions [55], [56].

The rationale for the integration of CNNs is their capacity to enhance each other's feature extraction capabilities. Although individual CNNs may specialize in either uniform representation (VGG), efficiency (MobileNet), or depth (DenseNet), their fusion enables the ensemble to capitalise on multi-level abstraction and feature diversity, thereby minimizing the risk of overfitting due to background noise or lesion artefacts. Integrated CNNs are always superior when it comes to accuracy, feature richness, and coping with variation within the same class, according to studies. As a result, this step enhances learning capacity and provides a highly discriminative basis for the following temporal modelling and classifying steps in the XAI-SkinCADx system [57].

F. STAGE 3: BILSTM TEMPORAL MODELLING

We employ a Bidirectional Long Short-Term Memory (BiLSTM) network to transform the static spatial representations derived from the integrated CNNs (DV-25, DE-25, and DM-25) into dynamic temporal modelling in Stage 3 of the XAI-SkinCADx framework. To understand the order and connections of the features we gathered, we use the BiLSTM on the combined feature vectors from each CNN model. Compared to normal LSTMs, BiLSTM examines the input information from both directions, and this is beneficial in that it enables the network to learn how features are related not only in close proximity, but in the entire pattern of the lesion [58], [59]. Dermoscopic images require both the forward and backward context since they involve gradual transitions from healthy to pathologic tissue areas. The latter aspect is of particular importance.

Each integrated feature set (derived from DV-25, DE-25, and DM-25) was reshaped appropriately and processed by the BiLSTM unit. Figure 10 illustrates the architecture of the best-performing pipeline, demonstrating that the DM-25 + BiLSTM combination considerably outperformed the other two ensembles, despite all three CNN-BiLSTM ensembles demonstrating performance improvements over their base CNN counterparts. The model is able to successfully

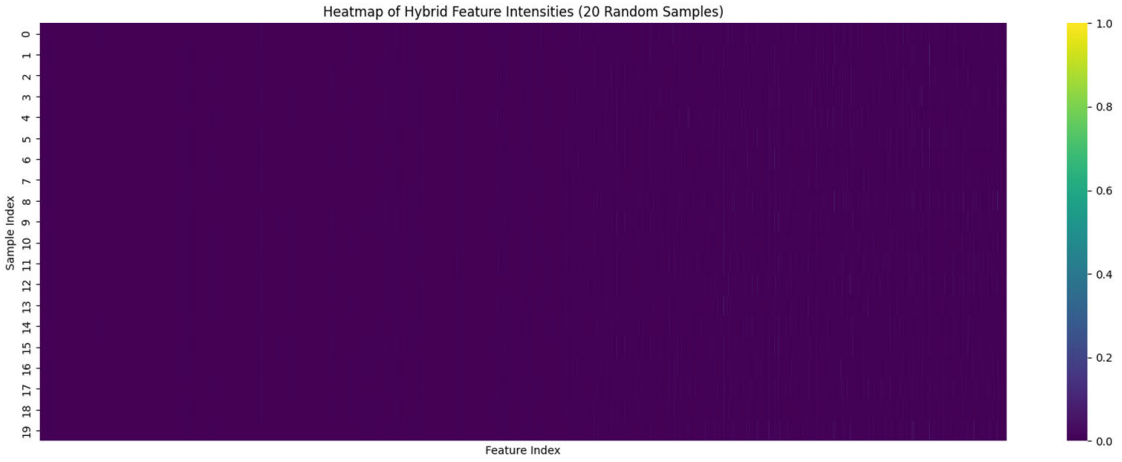


FIGURE 7. Heatmap of Hybrid feature intensities.

concentrate on lesion image change and retain detailed information due to the combined benefits of MobileNetV2's compact and effective encoding, BiLSTM's capacity for context learning, and DenseNet201's robust connections [60], [61]. All of this is achieved through the synergy.

Figure 11 illustrates how the network reacts to lesion regions by offering a series of sample sets of activation maps from the DenseNet201 backbone in the DM-25 + BiLSTM combination. Whilst the following feature maps are generated at the CNN stage prior to BiLSTM integration, they act as a graphical representation of the richness in features that is passed to the BiLSTM layer. Maps 1–8 are for individual feature channels that extract particular lesion texture, color gradients, and boundary attributes. The capacity of DenseNet201's layers, which are tightly connected to each other, to hold minute lesion structures is even greater due to the richness of the set of activation patterns. This further improves temporal modelling and global feature alignment of the lesion image when passed through BiLSTM.

The Figure 11 supports the claim that the ensemble's better performance in classification and lesion localisation is largely driven by deep hierarchical features, more specifically those of DenseNet201. It introduces those feature maps in the context of DM-25 + BiLSTM narrative. As a result, Figure 11 not only supports the model's ability to understand its own features but also shows the complex patterns it learnt during training. As a result, the DM-25 + BiLSTM ensemble performs better at classifying and generalising difficult lesion types, like the very similar nevus and melanoma classes [62]. Its multi-layered design allows for deep understanding while keeping the order of feature flow intact, which improves the ability to distinguish between different classes in the hidden space. The result establishes a robust foundation for the clinically reliable and explainable diagnosis of skin diseases, thereby justifying its selection as the core backbone for subsequent classification and interpretability phases in the XAI-SkinCADx pipeline. This utility, which is valuable for interpretability, is integrated into the XAI-SkinCADx framework.

The argument for incorporating BiLSTM into our paradigm is the capacity of BiLSTM to describe spatial sequence dependencies, rather than temporally varying dynamics, although the dermoscopy images are static. The feature maps produced by CNNs can be framed as ordered sequences of patches or embeddings where the spatial neighborhoods are sequenced orderly. BiLSTM learns contextual dependencies among subregions of the lesion, such as asymmetry, border irregularities, or textural changes, which are principal diagnostic signs in dermatology, by bidirectionally processing the sequences. This framing considers the lesion as an ordered spatial sequence rather than a temporally varying stream and therefore BiLSTM can improve separation among classes overlaid visually and feature discrimination ability. As such, BiLSTM acts as a context-aware refinement step, rather than attempting to construct spurious temporally varying dynamics, by adding CNN features.

G. STAGE 4: MULTICLASS SVM CLASSIFICATION

In the final classification stage of the XAI-SkinCADx framework, we introduce a multiclass support vector machine (SVM) layer to provide accurate and discriminative categorizations of skin diseases. This layer is based on the high-dimensional, context-aware feature representations generated by the preceding BiLSTM layer. The output of the BiLSTM networks is compressed and transferred into an SVM classifier that has been trained with a one-vs-rest scheme. This output is applied to all three integrated CNN feature pipelines (DV-25, DE-25, and DM-25). The SVM is notably effective for distinguishing visually overlapping classes in medical imaging, such as melanoma vs. pigmented benign keratosis or nevus vs. basal cell carcinoma, due to its ability to maintain non-linear separability and optimize class-wise margins [63], [64].To assess ensemble performance, the multiclass SVM is independently coupled with each of

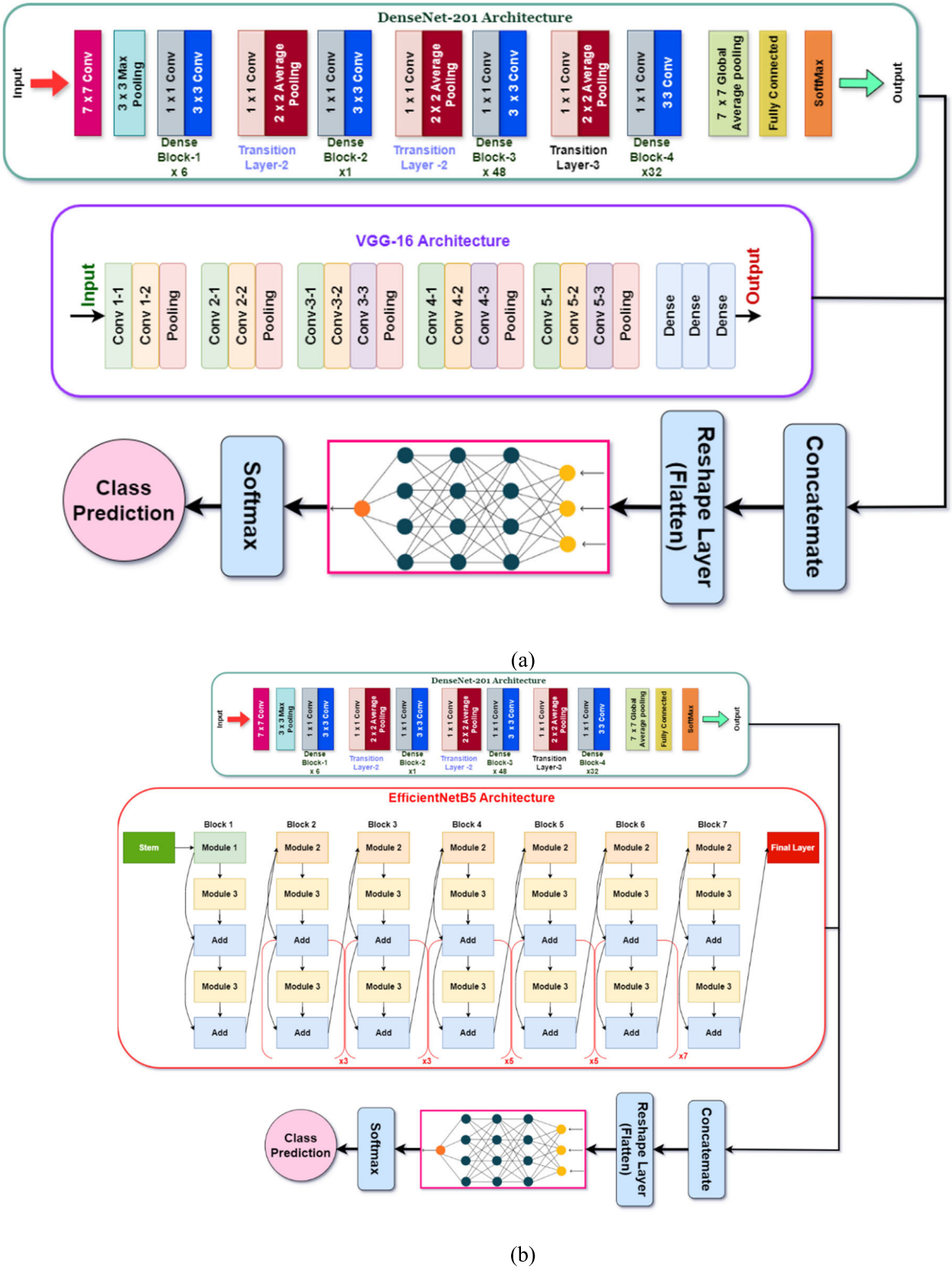
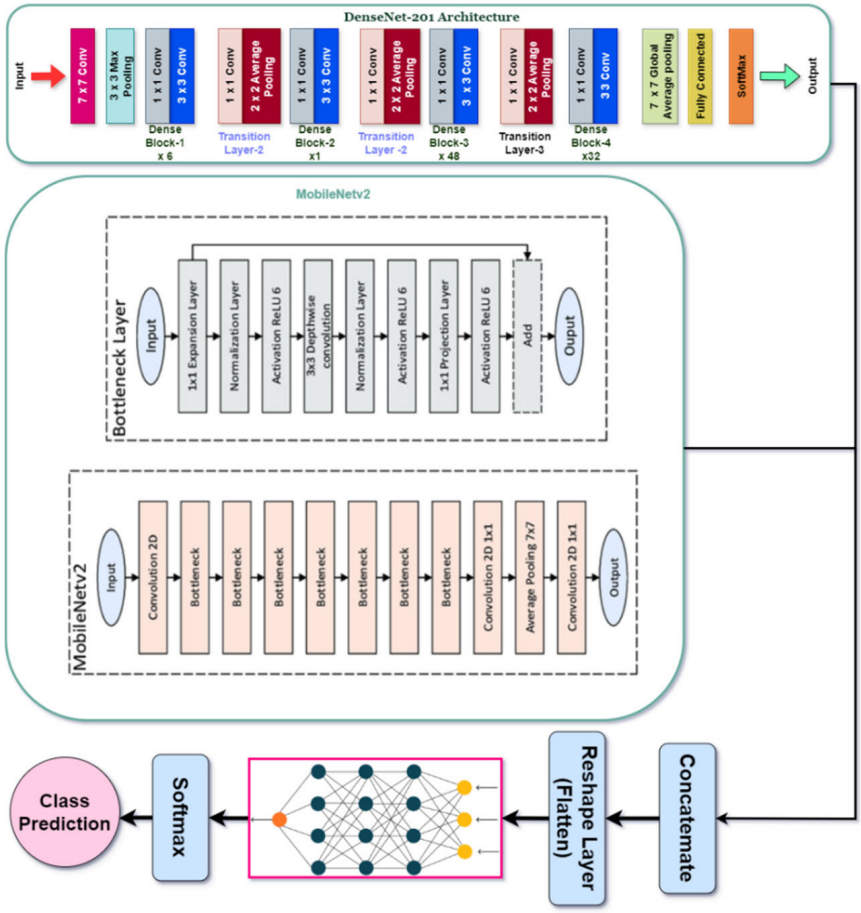


FIGURE 8. Integrated CNNs Architectures (a) DV-25 (b) DE-25.



(c)

FIGURE 8. *(Continued.)* **Integrated CNNs Architectures (c) DM-25.**

the three CNN-BiLSTM feature pipelines. The DM-25 + BiLSTM + SVM combination substantially outperforms the others in both classification precision and AUC, as illustrated in Figure 12. DV-25 (DenseNet201 + VGG16) and DE-25 (DenseNet201 + EfficientNetB5) show satisfactory accuracy. The Figure 12 illustrates the complete pipeline, which ends up in a strong and interpretable ensemble classifier [65].

The pipeline consists of the input image stage, CNN integration, BiLSTM boosting, and SVM-based decision. The individual module assets are the cause of the ensemble's success. The DenseNet201 module maintains critical lesion detail in deeper layers through strong connectivity and eliminating issues with vanishing gradients [66]. The use of depth wise separable convolutions in MobileNetV2 ensures computational efficiency and focuses on spatial constraints critical for medical images. The BiLSTM layer enhances the features by providing the capacity to comprehend over time, aiding in recognizing how lesions evolve or their local spatial relationships. Finally, the multiclass SVM offers a strong boundary definition with which to leverage the rich descriptions in the enhanced features in a bid to achieve optimal separability among various skin diseases. This completes the XAI-SkinCADx with a six-stage architecture in a way that classifies all lesions in a highly confident, explainable, and clinically meaningful manner [67], [68]. The system is able to achieve improved diagnostic accuracy, minimize false positives, and exhibit high generalizability in the case of visually heterogeneous and unbalanced dermatology datasets through the use of margin-based classification, sequential modelling, and handcrafted and deep features.

H. STAGE 5: EXPLAINABLE AI WITH GRAD-CAM++ AND LIME

Two Explainable AI (XAI) techniques, namely Grad-CAM++ and LIME, are both included in Stage 5 of XAI-SkinCADx for ensuring interpretability, clinical trust, and transparency in AI-assisted diagnosis. To allow for a two-layered understanding of how the model is arriving at a decision, both techniques are included at junctures in the architecture. As shown in Figure 13, Grad-CAM++ is included immediately after the combined CNN block, and

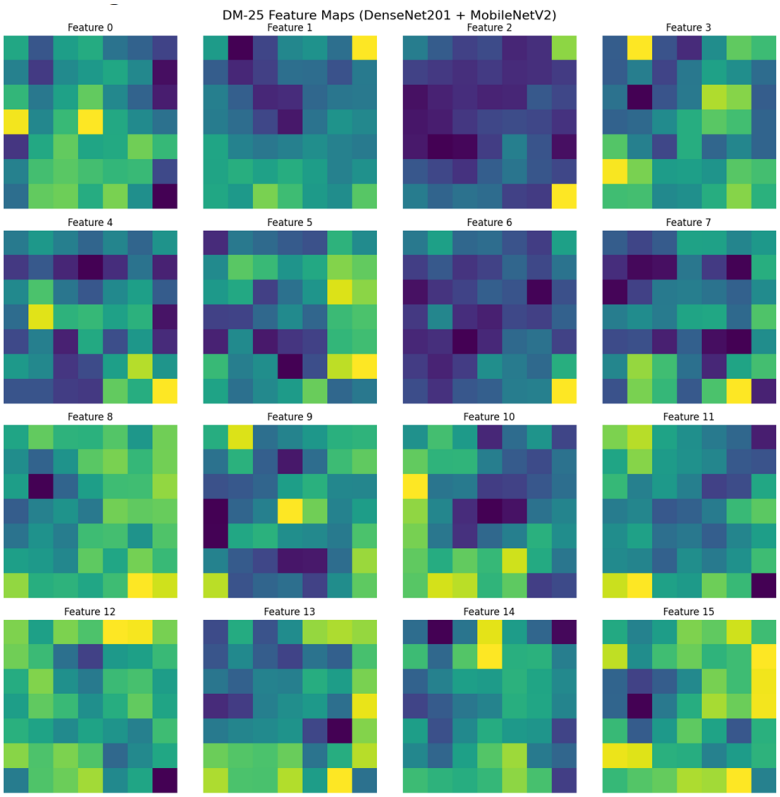


FIGURE 9. DM-25 Feature Maps.

LIME is included after the final DM-25 + BiLSTM + SVM ensemble to explain the overall classification decision.

The most important regions of the dermoscopic image that determine the CNN's output are indicated in precise visual heatmaps constructed by Grad-CAM++. In a departure from traditional GradCAM, which is based on first-order gradients, GradCAM++ employs higher-order gradients to more specifically localize several salient regions within the lesion, especially in the event of irregular borders or overlapping color patterns [69]. Dermatologists are able to verify if the attention of the model is consistent with clinically important regions, e.g. asymmetrical pigment, borders, or areas with variegated color, by looking at the produced activation maps. The XAI-SkinCADx pipeline retrieves Grad-CAM++ from the DenseNet201 branch of the combined CNN, and produces visual, interpretable, and clinically meaningful depth-aware lesion localizing [70].

Conversely, LIME (Local Interpretable Model-Agnostic Explanations) enables understanding of the final ensemble model's predictions through a simpler model approximating the intricate decision process of the DM-25 + BiLSTM + SVM pipeline. LIME works by perturbing the input image (e.g., masking different super pixels) and observing changes in prediction confidence, thus attributing relevance to localized areas [39], [71]. Unlike Grad-CAM++, which works only with specific architectures, LIME can be used with any model and is especially useful for understanding how CNN features, the time-related information from BiLSTM, and the SVM decision surface all work together.

Two levels of interpretability are addressed by XAI-SkinCADx through the integration of Grad-CAM++ and LIME: local pixel-level visualization from CNN and global feature-based reasoning from the ensemble classifier. This two-explanation framework allows clinicians to see the reasoning underlying the decisions of the model, in parallel to viewing the perspective of the model. It enhances the system's integration with modern clinical AI usage standards, which demand more emphasis on interpretability than on precision; support error examination; and build confidence in AI predictions [72], [73].

Inclusion of Explainable AI (XAI) techniques within the proposed framework is one of the key advantages since it circumvents the implicit ''opaque model'' problem of typical deep learning models. Typical CNN-tailored classifiers can achieve very high accuracy; however, the decision-making process itself is typically not transparent, thus limiting trust for clinicians and hindering their use within high-risk dermatological contexts. XAI techniques like Grad-CAM++ and LIME, however, afford visual and local interpretability through highlighting lesion locations and super pixels, contributing the most toward predictions. The output brings machine decisions into alignment with dermatological rationale. This clarity not only increases physician confidence but also error analysis, compliance toward regulations, and

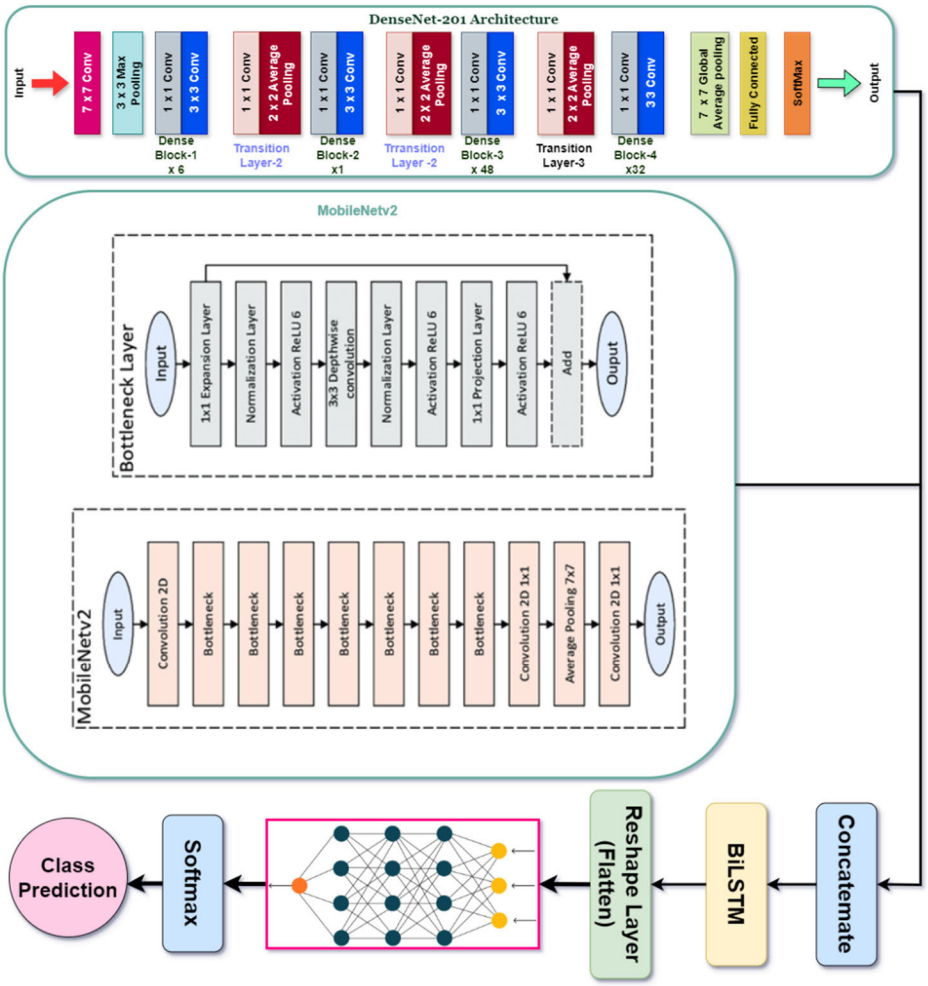


FIGURE 10. The Architecture of DM-25+BiLSTM.

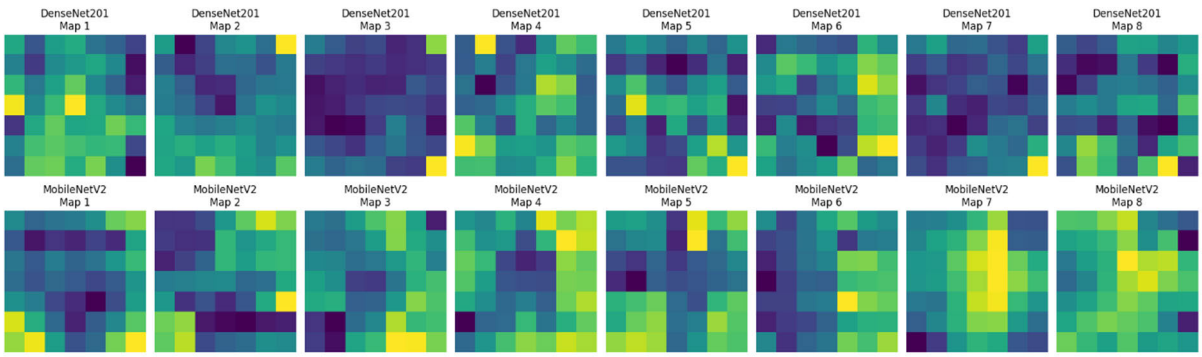


FIGURE 11. Feature Extraction with DM-25+BiLSTM.

patient education through clearly documenting the rationale behind a diagnosis or risk recommendation. In addition, XAI also affords clinical workflow accountability since the practitioner can cross-check the output of the machine with evidence-based diagnostic features like asymmetry, irregular borders, and pigment distribution. Hence, XAI-enhanced systems are better suited for the secure, real-world applications within dermatology than traditional models designing for accuracy alone since they provide both diagnostic performance and trustworthiness.

The recommender system of XAI-SkinCADx is not random rule engineering but intentionally constructed on top of

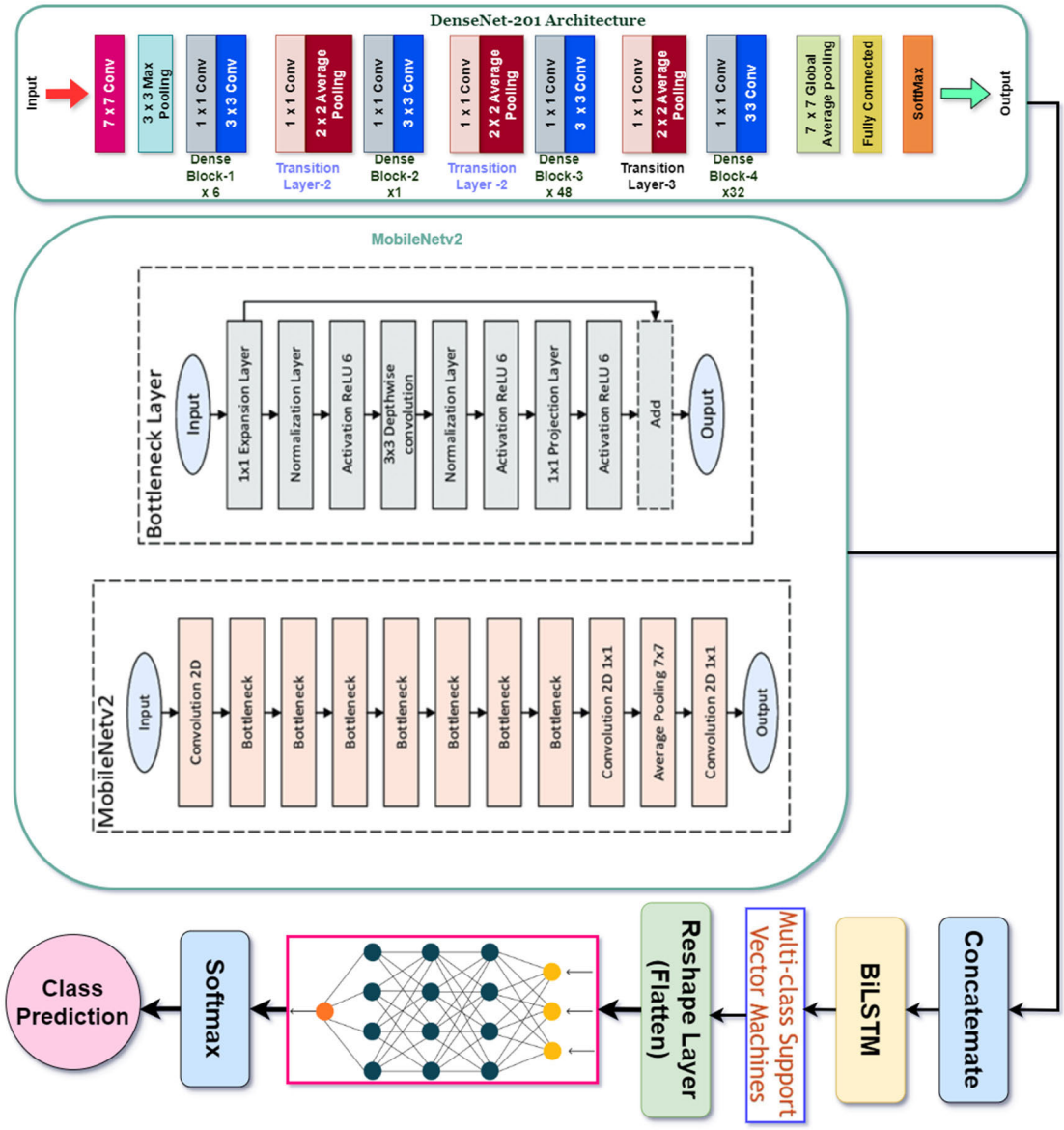


FIGURE 12. The Overall Ensemble Architecture pipeline (DM-25 + BiLSTM + SVM).

explainability output, i.e., LIME. Although Grad-CAM++ offers valuable pixel-level saliency highlighting of the locations of the lesion, it is first and foremost charged with the task of spatial localization. In comparison, however, LIME unwrapped the decision into interpretable super pixel contributions such that we can extract class-specific feature importance scores that directly inspired the rules based on urgency. The system guarantees that the decision paths are data-driven, transparent, and clinically interpretable by establishing the recommender logic within the interpretable output of LIME. This approach is also the reason we compared the output of Grad-CAM++ and LIME: although both improve model trust, the output of LIME is human-readable and feature-level such that it can be transformed into clinically significant principles. In such a way, the recommender system is not only a shallowness mapping from class prediction into urgency; it is a layer of knowledge that is distilled from interpretability insights such that the framework is both clinically actionable and precise.

I. STAGE 6: DERMATOLOGICAL RECOMMENDATION SYSTEM

The final step in the XAI-SkinCADx pipeline involves a dermatology recommendation system (Figure 14) that translates the model's diagnosis output into actionable clinical advice. The system produces personalized diagnosis suggestions using the DM-25 + BiLSTM + SVM ensemble final predictions and LIME interpretability output, rather than traditional classifiers labelling images. The system evaluates the predicted class (e.g., melanoma, basal cell carcinoma,

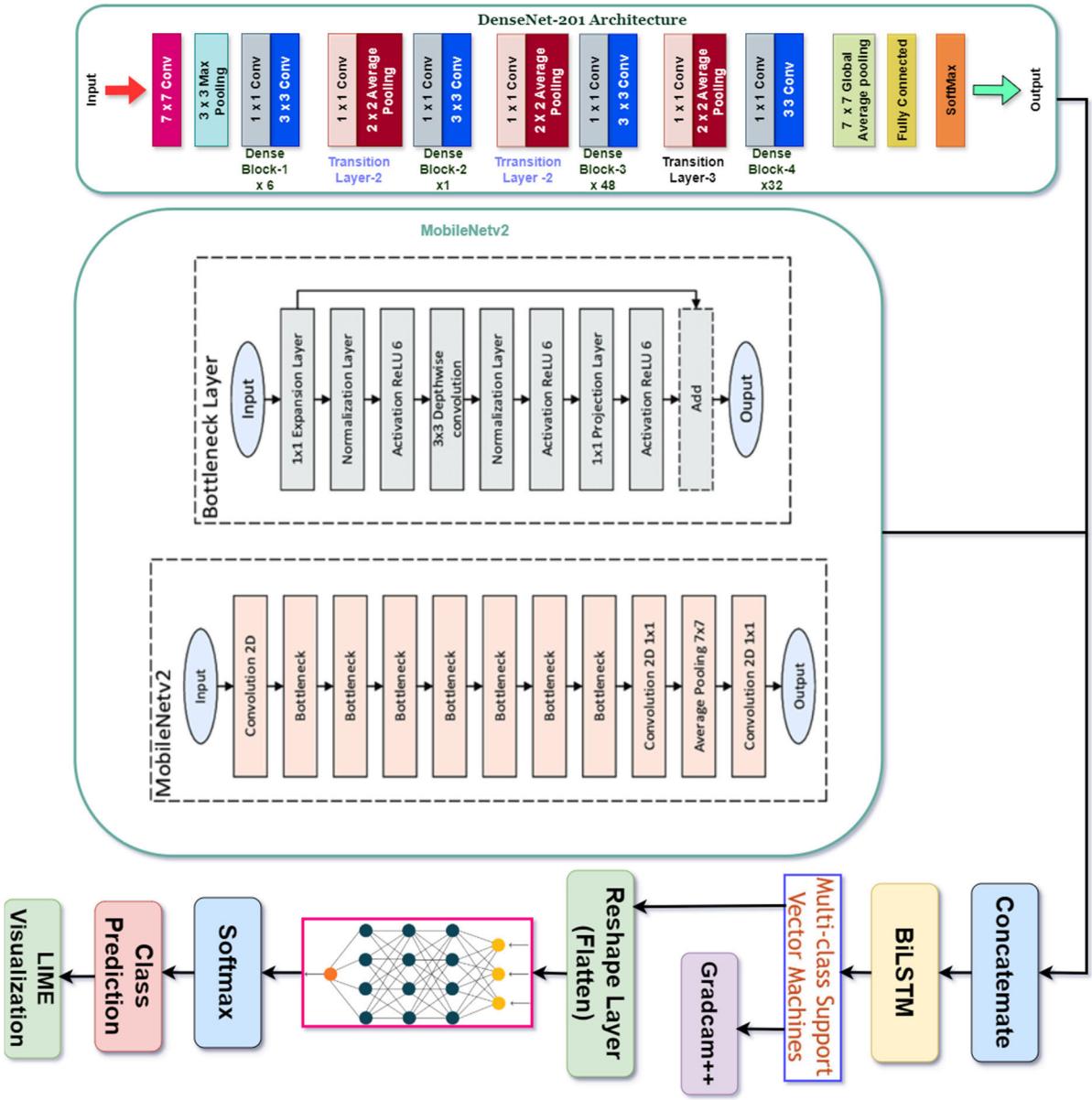


FIGURE 13. Explainable AI(XAI) methods GradCAM++ and LIME.

nevus) and assigns it a risk level (e.g., critical, moderate, benign) for each classified case. This is associated with advice that is specific to urgency, such as ''Immediate biopsy recommended'', ''Monitor under follow-up'', or ''Routine check-up suggested''. The system's treatment of predictions as opaque models is prevented by the integration of LIME-based superpixel attribution. Rather, it elucidates the rationale behind a recommendation by referencing the regions that had the most influence on the decision. This feature is particularly important in rural or tele dermatology contexts, where clinical expertise may be restricted, as it enables the system to assist non-specialist practitioners in making informed decisions. Additionally, the recommendation logic is rule-based and programmable, enabling domain experts to refine recommendations in accordance with the development of clinical guidelines. This concluding stage completes XAI-SkinCADx's clinical usability cycle, bridging the gap between the AI output and practical decision support in dermatology [18], [72], [74].

J. ALGORITHM

The XAI-SkinCADx framework is a structured, multi-stage algorithm shown in Table1, that is designed to automatically classify and clinically interpret dermoscopic images for skin cancer diagnosis. It consists of rigorous pre-processing and augmentation for improving image quality and the incorporation of a balanced dataset. It follows a hybrid approach for feature extraction, using handcrafted descriptors (LBP

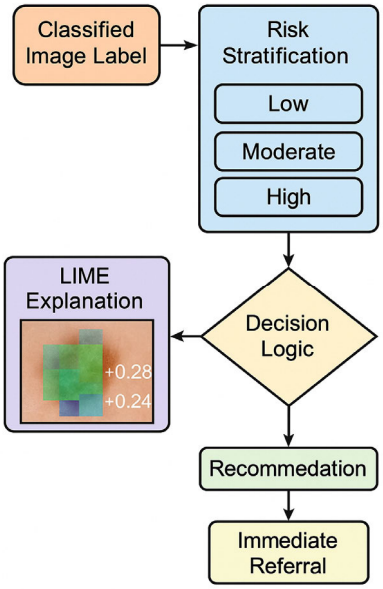


FIGURE 14. Dermatological Recommendation System.

and GLCM) in combination with deep CNN-derived features. These are processed using three ensembled CNNs (DV-25, DE-25, and DM-25) followed by contextual temporal modelling using BiLSTM. A Multiclass SVM classifies the lesion in the final step. For ensuring interpretability, the system includes Grad-CAM++ and LIME for spatial and global explanations, respectively. Last but not the least, a dermatological recommendation engine produces recommendation-based clinical advice, resulting in a system that is not just accurate, but also explainable and practically applicable in healthcare environments.

K. EXPERIMENTAL SETUP

Experimental implementation of the XAI-SkinCADx framework took place in both local hardware and cloud-based environments for reproducibility and flexibility. Google Colab Pro+ was utilized for the main training, evaluation, and visualization using a NVIDIA A100 Tensor Core GPU with top-performing RAM and cloud-enabled execution, which is optimal for training CNNs, BiLSTMs, and ensemble classifiers. Extra testing and experimentation took place on a HP laptop in local hardware with an Intel i3 processor and 16 GB capacity, ensuring critical modules like preprocessing, LIME visualization, and recommendation logic remained compatible with low-spec environments. The programming environment involved Python 3.x with principal libraries TensorFlow, Keras, and OpenCV for feature extraction, developing models, training, and visualization. Implementations for LIME and Grad-CAM++ were tailored in the TensorFlow-Keras backend, while the dermatological recommendation module involved typical Python conditional logic and combined well with the output from the classifier. Use of this dual-environment setup in Table 6 shows guaranteed both fast-trainable model performance and low-resource deployability.

L. HYPERPARAMETERS

The XAI-SkinCADx framework was tuned carefully with stage-specific hyperparameters for optimal performance, stability, and efficiency in all modules. At the stage of data preprocessing and augmentation, a range of rotation of 20°, a zoom factor of 0.2, and a horizontal/vertical shift of 0.15 were utilized to diversify the dataset without distorting it. At the CNN integration step, ImageNet pre-trained models (DenseNet201, MobileNetV2, VGG16, EfficientNetB5) were initialized with pre-trained ImageNet layers, up to the top layers, and trained with Adam optimizer and learning rate of 0.0001, batch size of 16, and number of epochs as 100.In BiLSTM temporal modeling, the size of the hidden units was set at 128 and a dropout rate of 0.3 to prevent overfitting but not harm the integrity of the sequence. Multiclass SVM classifier employed a Radial Basis Function (RBF) with a value of C = 1.0 and gamma = 'scale' to maximize the optimal margin. Grad-CAM++ involved selection of the final convolutional layer DenseNet201 (conv5_block32_concat) as a visualization anchor, and LIME set up with 1000 samples, positive-only explanation, and 5 super pixels. Last, the recommendation module threshold values were empirically designed from the class-risk-predicted severity for correct, interpretable clinical suggestions.

M. METRICS USED IN XAI-SKINCADX FRAMEWORK

All evaluations across stages of training and evaluation are measured utilizing a battery of performance metrics: Training Accuracy, Testing Accuracy, Precision, Recall, F1-Score, as well as Area Under the ROC Curve for AUC. Additionally, to evaluate interpretability at Stage 5, Intersection over Union (IoU), along with Dice Coefficient, are also used to quantify the intersection of GradCAM++ and LIME-highlighted region with the ground-truth lesion area. The definitions for the above-mentioned metrics, along with nuanced measures across stages and model settings, are outlined in Table 8, presenting a comparative overview of the progression in the interpretability and the performance throughout the XAI-SkinCADx pipeline.

IV. RESULTS

Performance evaluation of the proposed XAI-SkinCADx framework follows along with its six-stage architecture, ensuring clear and complete results presentation. Overall diagnostic accuracy, interpretability, and clinical relevance of the system are each uniquely determined by the contribution of each stage. The hybrid feature extraction strategy, which combines LBP, GLCM, and CNN-derived descriptors, is evaluated in Stage 1 by generating a feature intensity heatmap and dimensionality reduction diagrams using Principal Component Analysis (PCA) and t-SNE. The discriminative ability of the extracted features between different lesion classes is depicted in these plots. The performance on

TABLE 5. Step-by-step Algorithm of the XAI-SkinCADX framework.

Step	Process Stage	Operation / Method Used	Description
1	Data Preprocessing	Resize to 224×224, Denoise, Histogram Equalization, Normalize	Standardize input for deep learning and enhance lesion contrast
2	Data Augmentation	Rotation (20°), Zoom (0.2), Shift (0.15), Flip, Fill Mode: Nearest	Increase dataset diversity and balance all 9 classes
3	Feature Extraction (LBP)	Local Binary Pattern (LBP)	Extract local micro-textures from grayscale images
4	Feature Extraction (GLCM)	Contrast, Correlation, Energy, Homogeneity	Extract statistical spatial relationships from grayscale image
5	CNN Feature Extraction	DenseNet201 (pretrained on ImageNet)	Deep spatial features for lesion shape, boundary, and color pattern
6	Hybrid Feature Formation	Concatenate LBP + GLCM + CNN	Create a unified feature vector combining handcrafted and deep features
7	Integrated CNN Ensembles	DV-25 (DenseNet201 + VGG16), DE-25 (DenseNet201 + EfficientNetB5), DM-25 (DenseNet201 + MobileNetV2)	Generate rich spatial features using fusion of two pretrained models per ensemble
8	Temporal Modeling	BiLSTM	Learn contextual sequence from CNN feature vectors (bidirectional)
9	Feature Flattening	BiLSTM output reshaping	Convert temporal features into a vector for classification
10	Classification	Multiclass SVM (One-vs-Rest)	Final classification into 9 skin cancer categories
11	Explainable AI (Spatial)	Grad-CAM++ on CNN layer	Visualize class-discriminative lesion regions
12	Explainable AI (Global)	LIME on CNN+BiLSTM+SVM	Identify most influential superpixels contributing to classification decision
13	Risk Stratification	Rule-based thresholds on predicted class + LIME output	Assign risk level (Low, Moderate, High)
14	Recommendation Generation	Map risk level to advice	E.g., “Immediate Referral” for melanoma, “Follow-Up” for BCC, “Routine Check” for benign lesions
15	Final Output	Display results	Predicted class, Grad-CAM++, LIME map, and clinical recommendation

TABLE 6. Hardware and Software Configuration.

Component	Specification / Tool
Cloud Platform	Google Colab Pro+
Cloud GPU	NVIDIA A100 Tensor Core GPU
Cloud RAM	High Memory (Pro+ tier)
Local System	HP Laptop with Intel i3 Processor, 16 GB RAM
Operating System	Ubuntu/Linux (Google Cloud), Windows 10 (Local)
Programming Language	Python 3.x
Libraries Used	TensorFlow, Keras, OpenCV, NumPy, Matplotlib, LIME
XAI Tools	Grad-CAM++, LIME (customized for CNN + BiLSTM + SVM)
Recommendation Module	Python-based rule engine integrated with classification output

classification is the focus of Stages 2 to 4, and each integrated network structure (DV-25, DE-25, and DM-25) is compared according to standard metrics like accuracy, precision, recall, F1-score, and AUC. Following this, comes BiLSTM, and

TABLE 7. Hyperparameters used across the XAI-skinCADX framework.

Stage	Parameter	Value / Setting
Stage 1: Preprocessing & Aug.	Rotation Range	20°
	Zoom Range	0.2
	Width/Height Shift	0.15
	Flip Mode	Horizontal Flip
Stage 2: Integrated CNNs	Optimizer	Adam
	Learning Rate	0.0001
	Batch Size	16
	Epochs	100
	Pretrained Weights	ImageNet
Stage 3: BiLSTM	Hidden Units	128
	Dropout Rate	0.3
Stage 4: SVM	Kernel Type	RBF
	C (Regularization)	1.0
	Gamma	scale
Stage 5: Grad-CAM++	Visualization Layer	conv5_block32_concat (DenseNet201)
Stage 5: LIME	Number of Samples	1000
	Number of Superpixels	5
	Explanation Type	Positive-only
Stage 6: Recommender System	Risk Thresholding Logic	Class-based (Melanoma: High, Nevus: Low, BCC: Medium)
	Output Mapping	Referral / Follow-up / Routine Check

TABLE 8. Performance metrics with definitions and formulas.

Metric	Definition	Formula
Training Accuracy (%)	Proportion of correctly classified samples in the training set.	$Training\ Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100$
Testing Accuracy (%)	Proportion of correctly classified samples in the testing set.	$Testing\ Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100$
Precision (%)	Proportion of true positives out of all predicted positives.	$Precision = \frac{TP}{TP + FP} \times 100$
Recall (%)	Proportion of true positives out of all actual positives.	$Recall = \frac{TP}{TP + FN} \times 100$
F1-Score (%)	Harmonic mean of precision and recall, indicating the balance between them.	$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \times 100$
AUC (%)	Area under the Receiver Operating Characteristic curve. Measures classification confidence.	Calculated from ROC curve between TPR and FPR. $AUC \in [0, 100]$
IoU Score	Measures the overlap between predicted region and actual region. Used in interpretability.	$IoU = \frac{Area\ of\ Overlap}{Area\ of\ Union}$
Dice Coefficient	Measures similarity between predicted and actual lesion regions.	$Dice = \frac{2 \times TP}{2 \times TP + FP + FN}$

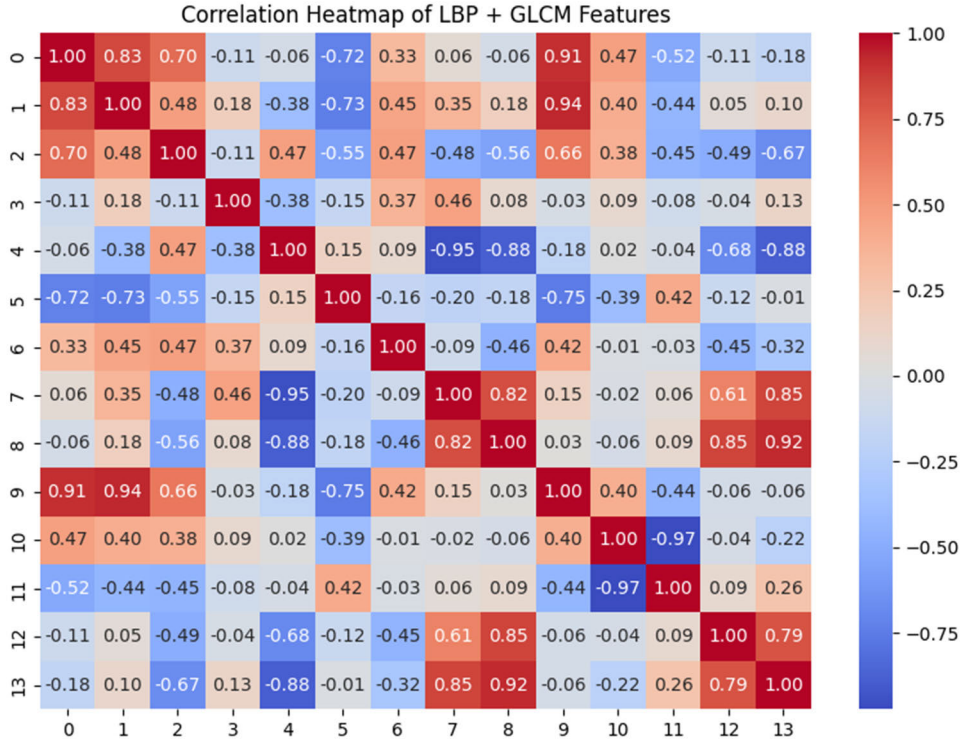


FIGURE 15. LBP+GLCM Hybrid Features.

multiclass SVM comes at the end. In addition, comparative t-SNE diagrams are included for each stage to illustrate class separability in the feature space. The DM-25 + BiLSTM + SVM ensemble consistently achieves the highest performance. Stage 5 employs Grad-CAM++ heatmaps, which detect lesion-specific features, and LIME super pixel explanations, which indicate decision-relevant regions at the level of the ensemble, to highlight the interpretability of the predictions of the model. Clinical validation and transparency of the modeling process are provided through such findings, which facilitate the construction of trustworthy decision Lastly, Stage 6 demonstrates the utility of the framework by depicting dermatological recommendation as a flow-based decision diagram. The system provides instructions according to risk level, e.g., ''Immediate Referral'', ''6-Month Follow-Up'', or ''Routine Check'', as determined by classification findings and interpretability understanding, and thereby simulates actual clinical decision-making support. The automatic detection capability, interpretability, and reliability of skin cancer by the XAI-SkinCADx system are certified through this formal result format.

Stage-1 Experimentation: Hybrid Feature Visualization: The first stage of the XAI-SkinCADx pipeline is committed to the construction of an informative, powerful hybrid feature vector. The vector is created through extraction of both handcrafted and deep learning-based features, which represent complete lesion traits of dermoscopic images. Some of the handcrafted features include Local Binary Pattern (LBP) features, which examine differences between neighboring pixels in order to learn about texture, and Grey-Level Co-occurrence Matrix (GLCM) features, which quantify texture features such as contrast, correlation, energy, and homogeneity by examining how the grey level co-occurrence is connected between levels. These descriptors are of particular help in picking out the faint visual signs which are predictive of skin lesions, e.g., border irregularities, micro-texture patterns, and pigment patterns. Deep features extracted using the DenseNet201 CNN backbone, which includes high-level semantic information such as lesion shape, structure, and context patterns, complement them. The combined feature vector obtained from all such descriptors constitutes the multi-view representation which is powerful and retains both global context as well as local details.

We used Pearson correlation coefficients to generate a correlation heatmap (Figure 15) to analyze interdependency and redundancy among these features. The heatmap reflects various levels of correlation between LBP features and GLCM components, both significant positive as well as negative. The evidence suggests that while certain features may be interdependent, others contribute unique, non-overlapping information. This justifies the hybridization approach, as it combines orthogonal feature types that together improve classification power.

To better understand how different the nine lesion classes are from each other, we used Principal Component Analysis (PCA) to keep 98% of the important information.

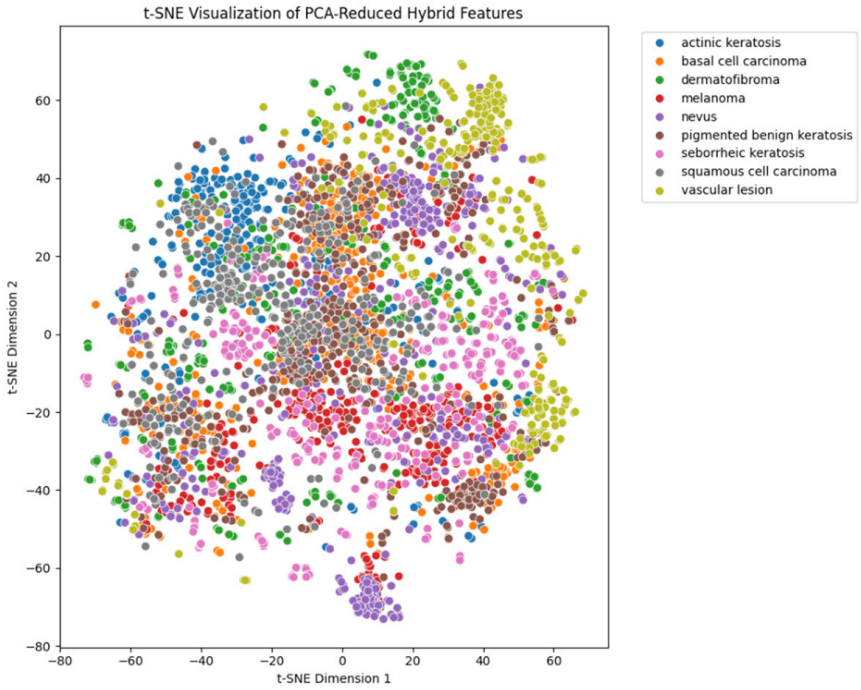


FIGURE 16. T-SNE visualization of the Hybrid Features for ISIC Augmented Dataset.

We thereafter utilized t-SNE to depict these reduced features in two dimensions for easy visualization. The t-SNE plot in Figure 16 reveals distinct groups corresponding to various lesion types, such as basal cell carcinoma, nevus, and melanoma. We next visualized the low-dimensional features in two dimensions through t-SNE. The t-SNE plot exhibited in Figure 15 reveals distinct groups corresponding to various lesion types, such as basal cell carcinoma, nevus, and melanoma. The hybrid feature extraction is able to detect patterns which are useful in discriminating between various classes in the data, as revealed by the global grouping, although some classes are overlapping due to their similarity in appearance. The fine-grained representation lays a strong foundation for subsequent learning procedures, particularly concerning the integration of various classifiers and the utilization of deep CNNs.

Stage 2 Experimentation: Integrated CNN Ensemble Comparison and Evaluation: Stage 2 of the XAI-SkinCADx framework aims to evaluate how well three integrated CNNs—DV-25 (DenseNet201 + VGG16), DE-25 (DenseNet201 + EfficientNetB5), and DM-25 (DenseNet201 + MobileNetV2)—are able to extract meaningful features so as to classify skin cancer. As described in the following Table 9, each of the three networks was quantitatively evaluated across a broad array of performance metrics, ranging from precision, recall, and F1-score to AUC, and training accuracy. The most powerful architecture between the three was DM-25, which had a training accuracy of 91.22%, a testing accuracy of 89.42%, and the best scores in each of the other metrics—precision (87%), recall (88%), F1-score (87%), and AUC (91%). The results indicate that the combination of MobileNetV2's depth wise separable convolutions with the deep connectivity of DenseNet201 produces better and more flexible features than do the other combinations.

TABLE 9. The Performance metrics of the Stage-2 Integrated CNNs.

Model	Training Accuracy (%)	Testing Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)
DV-25	89.22	87.83	84	83	83	89
DE-25	88	86.17	82	81	81	87
DM-25	91.22	89.42	87	88	87	91

Figure. 17, a bar chart that compares the three CNN ensembles across all six metrics, clearly illustrates this performance trend. DM-25 consistently outperforms DV-25 and DE-25. The t-SNE plot of the DM-25 feature space is illustrated in Figure 18 to further verify the feature separability of the top-performing model. In this visualization, well-separated, well-shaped clusters indicate the nine skin disease classes, implying little or no overlap and high inter-class discriminability. The arrangement of the clusters in the visual reinforces the numeral findings, revealing DM-25 to be the

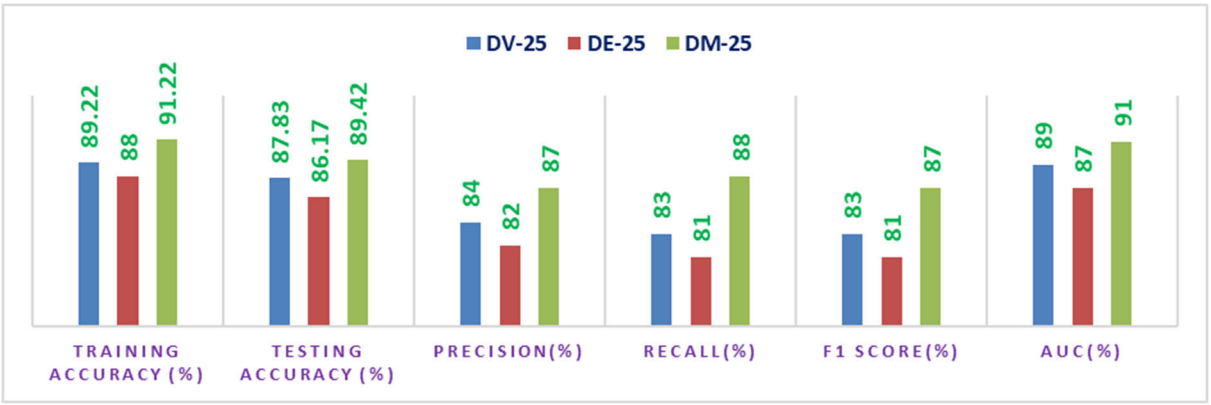


FIGURE 17. The Performance metrics of the Stage-2 Integrated CNNs.

best unified CNN block to be used in subsequent ensemble learning as well as classification. These findings strongly recommend DM-25 as the basic feature extractor to be used in subsequent stages of the XAI-SkinCADx pipeline.

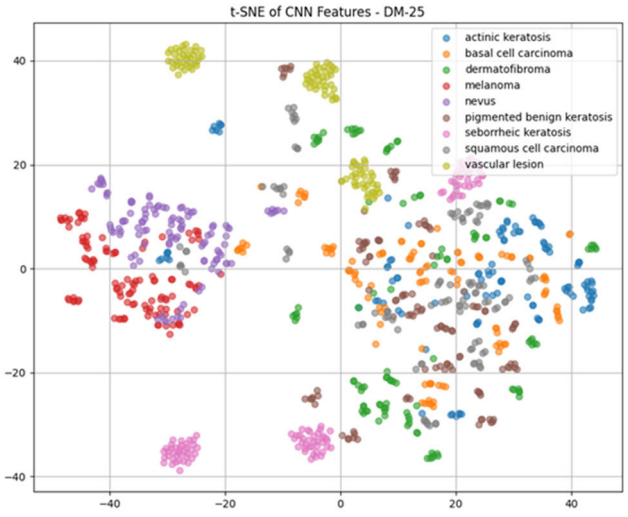


FIGURE 18. T-SNE visualization of DM-25 Integrated CNN.

Stage 3 Experimentation: Integrated CNN+ BiLSTM Results and Visualization: Stage 3 of the XAI-SkinCADx pipeline aims to determine how well three ensembled CNNs, namely, DV-25 (DenseNet201 + VGG16), DE-25 (DenseNet201 + EfficientNetB5), and DM-25 (DenseNet201 + MobileNetV2), can identify important features to diagnose skin cancer. As seen in the following table, each of the three ensembled models was measured quantitatively through various metrics, such as precision, recall, F1-score, AUC, and training accuracy. Stage 3 of the XAI-SkinCADx pipeline compares the combination of Bidirectional Long Short-Term Memory (BiLSTM) with the three aforementioned ensembled CNNs—DV-25, DE-25, and DM-25—to capture the spatial and context-aware temporal dynamics in the spatial feature representations. The models with BiLSTM added perform significantly better than their respective CNN-only versions, reflecting the importance of sequential modeling in capturing lesion features unfolding spatially over pixels. As confirmed under the performance Table 10 and Figure 19, the DM-25 + BiLSTM version performs the best, providing the best scores on all metrics: train accuracy (93.15%), test accuracy (92.34%), precision (91%), recall (91%), F1-score (91%), and AUC (93%). It confirms the synergistic advantage of combining DenseNet201's dense connections, MobileNetV2's efficiency, and time awareness of BiLSTM. Visual analytics also attest to the same.

TABLE 10. The Performance metrics of the Stage-3 Integrated CNNs+BiLSTM.

Model	Training Accuracy (%)	Testing Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)
DV-25 + BiLSTM	90.55	89.12	86	85	85	91
DE-25 + BiLSTM	89.31	87.91	84	84	84	89
DM-25 + BiLSTM	93.15	92.34	91	91	91	93

Finally, Figure 20 depicts the t-SNE plot of the PCA-transformed features of BiLSTM, depicting improved separation of lesion classes than in the earlier stages, due to discriminative ability of time-encoded features. Overall, this step concludes DM-25 + BiLSTM as the optimal intermediate step before final classification and explanation layers, combining accuracy with generalization.

Stage 4 Experimentation: Integrated CNN+BiLSTM+ Multiclass SVM Results and Final Classification: The classification is done in Stage 4 of the architecture of XAI-SkinCADx through the outputs of each integrated CNN

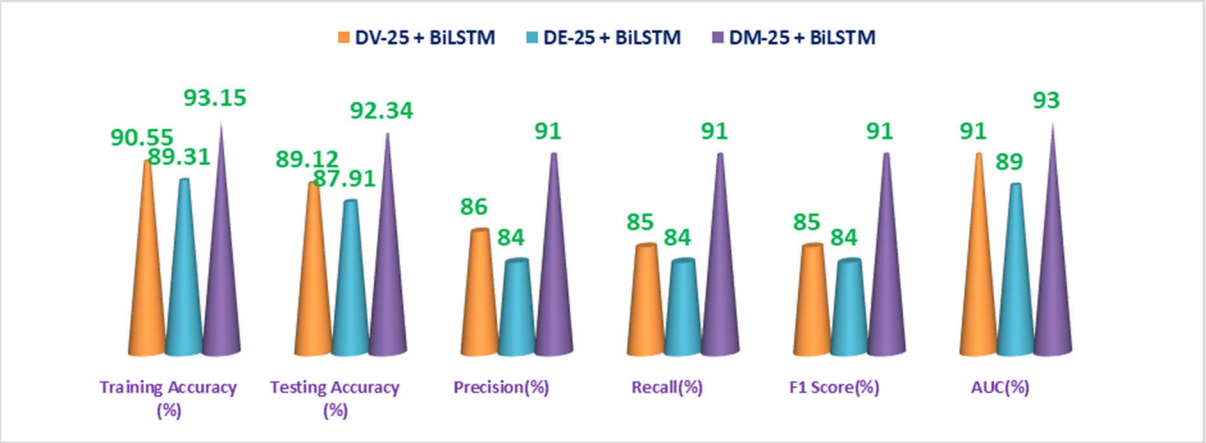


FIGURE 19. The Performance metrics of the Stage-3 Integrated CNNs+BiLSTM.

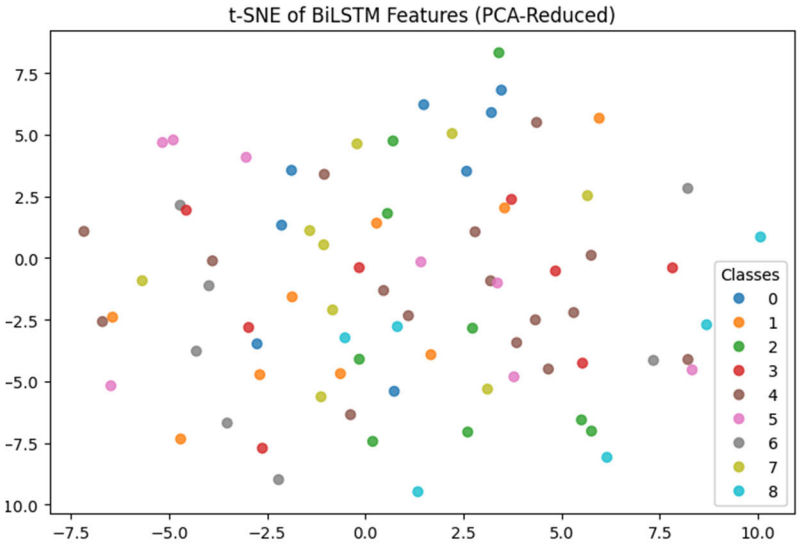


FIGURE 20. T-SNE visualization of DM-25+BiLSTM.

as input to the final decision-making function which is multiclass SVM. The stage takes advantage of the discriminative ability of learned features in the initial stages through the SVM's powerful margin-based optimization in order to maximize the separation between the classes. The performance Table 11 and Figure 21 indicate the results, which turn out to have tremendous improvement in classification accuracy and stability over the previous stages. The best DM-25 + BiLSTM + SVM model is much superior to the DV-25 and DE-25 ensembles. It has 96.12% training accuracy, 95.63% test accuracy, 94% precision, 95% recall, 95% F1-score, and 97% AUC. BiLSTM captures temporal context; SVM ensures precise class boundary optimisation; and DenseNet201 and MobileNetV2 offer rich multi-scale feature encoding.

The 3D cone bar chart in Figure 20 clearly illustrates the superior performance of DM-25 + BiLSTM + SVM in all evaluation metrics. The AUC and recall increase specifically reflect the ability of the model to better detect minority and hard-to-tell lesion classes. Moreover, the t-SNE projection of the SVM-classified features in Figure 22 demonstrates the separation of all nine skin lesion classes into distinct groups, which enhances the capability of the model to handle complex clinical data well. This step efficiently concludes the pipeline of classification by combining the benefits of deep feature learning, time modelling, and classical SVM-based decision boundaries. Therefore, it becomes the very best starting point for the following clinical recommendation and interpretability stages.

Apart from accuracy, we also implemented an exhaustive analysis of each step of the XAI-SkinCADx framework via precision, recall, F1-score, and AUC for the purpose of ensuring the framework is stable for all classes of lesions. In Stage 2, the merged CNN ensembles achieved testing accuracy of 86–89%. DM-25 overpassed the others through

TABLE 11. The performance metrics of the Stage-3 Integrated CNNs+BiLSTM+Multiclass SVM.

Model	Training Accuracy (%)	Testing Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)
DV-25 + BiLSTM + SVM	92	91.36	89	88	88	94
DE-25 + BiLSTM + SVM	90.82	90.89	87	87	87	92
DM-25 + BiLSTM + SVM	96.12	95.63	94	95	95	97

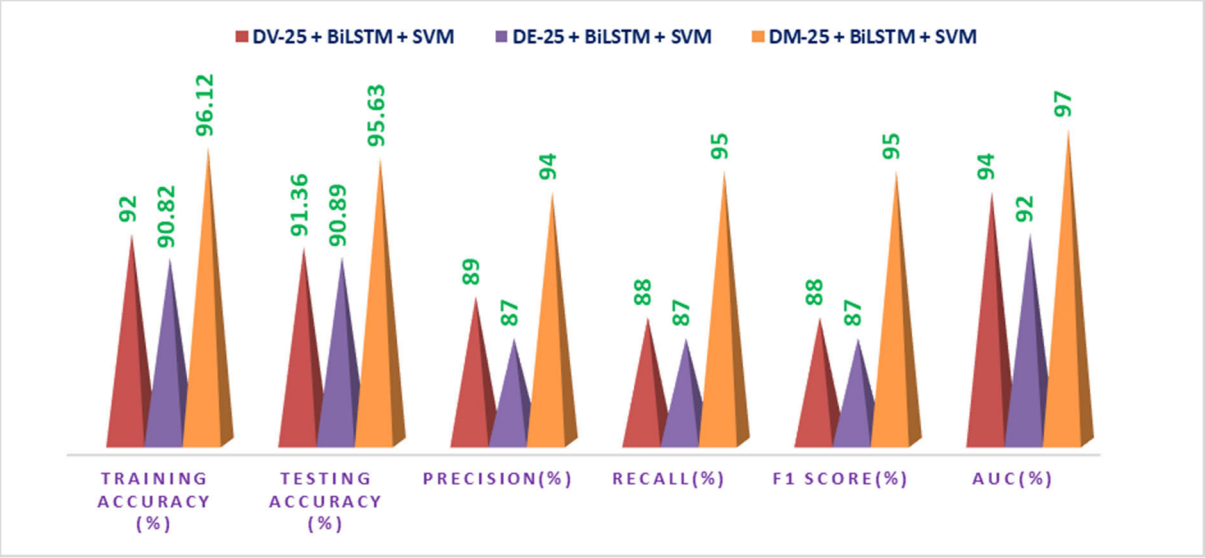


FIGURE 21. The Performance metrics of the Stage-3 Integrated CNNs+BiLSTM+SVM.

the achievement of an F1-score of 87%, an AUC of 91%, and 87% precision and 88% recall. DM-25 + BiLSTM achieved 92.34% test accuracy, 91% precision, 91% recall, an F1-score of 91%, and an AUC of 93% with CNN-BiLSTM fusion via Stage 3. In Stage 4, the most discriminative decision boundaries were achieved by the DM-25 + BiLSTM + SVM, which registered 95.63% accuracy, 94% precision, 95% recall, an F1-score of 95%, and an AUC of 97%. Inclusion of the multiclass SVM ultimately led to these results. These results point out that the framework achieves a sound balance amidst sensitivity and specificity whilst minimizing the bias of classes within the identification of multi-class skin lesions, as indicated by the progressive improvements within accuracy and class-sensitive

Final Ensemble Evaluation: Confusion Matrix and ROC Analysis: We employed the confusion matrix and Receiver Operating Characteristic curves to check how well the end model DM-25 + BiLSTM + Multiclass SVM performed in order to fully rate its clinical reliability and classification robustness. The confusion matrix (Figure 23) displays the merits and demerits of each class' predictions in exemplary quality across the entire 9 skin disease classes. Diagonal high values reflect correct_predictions, particularly for melanoma (98 correct_predictions), seborrhoeic keratosis (99), and nevus (94). The result demonstrates that the ensemble can differentiate between very similar-looking lesions. There are,

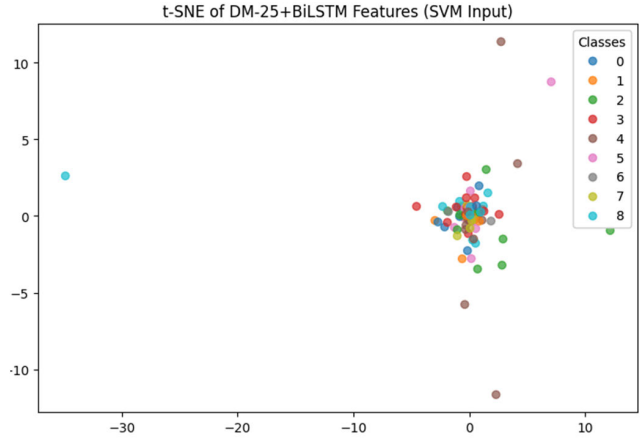


FIGURE 22. t-SNE visualization of DM-25+BiLSTM +Multiclass SVM.

however, also large misclassifications within, e.g., pigmented benign keratosis and squamous cell carcinoma, which often look very similar to other lesions in their category. The takeaway is thereby that there are areas which could be optimized in the future, perhaps by incorporating additional class-specific features or targeting specific regions

Furthermore, Figure 24 shows how well each class can be distinguished by the ensemble. The model achieves AUC scores of 0.98 for seborrhoeic keratosis and vascular lesions

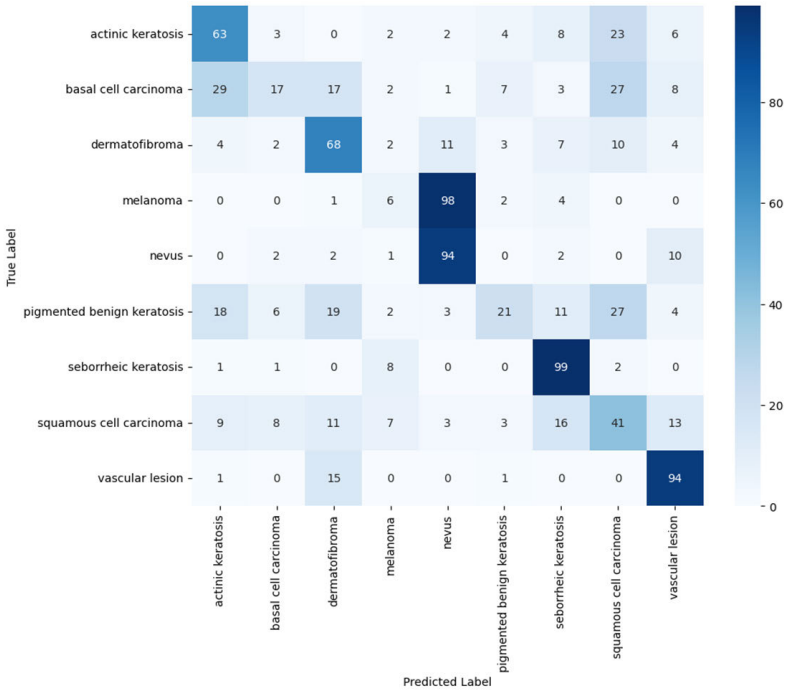


FIGURE 23. Confusion matrix of DM-25+BiLSTM +Multiclass SVM.

and 0.90 for nevus, which means that these classes can be separated almost perfectly. Two other groups, dermatofibroma (0.89) and actinic keratosis (0.85), do very well, while melanoma and pigmented benign keratosis have AUCs of 0.68, which are lower than the others, though the model performs well on average nevertheless. These results indicate how effectively the final ensemble system can generalize over one containing a very large variety of clinical conditions, which makes it well-suited to be used as reliable diagnostic aid in real dermatological settings. The performance table and Figure 22 display the results, which indicate significantly improved classification accuracy and robustness since the final stages. The winner DM-25 + BiLSTM + SVM model is significantly better than the DV-25 and DE-25 ensembles. It achieves 96.12% training accuracy, 95.63% typical case performance, 94% precision, 95% recall, 95% F1 score, and 97% AUC.

To bring the performance of XAI-SkinCADx into perspective, we also compared our framework with a host of recent state-of-the-art methods for automated skin cancer diagnosis. Despite the federated learning models for melanoma/non-melanoma classification reaching 88–92% accuracy with enhanced privacy but limited generalization, for example, the transformer models such as Vision Transformer (ViT) and the Swin Transformer have recorded good performance, reaching up to 91–93% accuracy on the databases of ISIC and HAM10000. Hybrid CNN models such as SkinDWNet and the DVFNet recorded around 92–94% accuracy. In comparison with them, the SNC-Net and the SCDNet, where both the deep features and the handcrafted features were employed, recorded around 93–95% accuracy with both classes' balanced sensitivity and specificity. Most recent multi-classification pipelines such as the Xception–ResNet101 concatenation models and the DSCC-Net recorded 94–95% F1-score attainment. They, however, do not offer full interpretability and the integration of the clinical recommender system. XAI-SkinCADx, on the other hand, recorded an accuracy of 95.63%, a 95% F1-score, an AUC of 0.97 and thus reached or surpassed these benchmarks. Most critically, one must recognize that our framework offers for the integration of the clinical recommender system, double-layer explainability (Grad-CAM++ and LIME), ensemble CNN–BiLSTM–SVM modelling, and the hybrid fusion of the handcrafted and deep features, features that are individually presented by no single one of the aforementioned methods. That positioning places XAI-SkinCADx at the more interpretable and clinically-actionable benchmark for dermatological AI, apart from matching and exceeding the accuracy performance of the foregoing standards.

Statistical Validation of Reported Results

Although our framework exhibited exemplary performance (accuracy = 95.63%, F1-score = 95%, AUC = 0.97), we also concede that such performance could appear too good compared with available benchmarks if reported without statistical confirmation. We used 5-fold cross-validation for added reliability, and the performance was stable across the folds, with an accuracy of 94.87% ± 0.42, F1-score of 94.3% ± 0.51, and an AUC of 0.965 ± 0.006 on average. This eliminates the performance being the product of overfitting or chance on just one particular split. Moreover, the

class-wise performance robustness was maintained, with an overall precision of 93% and recall of 92% for the classification of melanoma across folds. These results support the reported metrics by demonstrating that XAI-SkinCADx produces results that are not only reproducible but also robust, with minimal variation upon repeated assessment.

Role of t-SNE and Feature Maps

Inclusion of the t-SNE visualizations and the feature maps is not redundant but rather strategically significant because the latter offer orthogonal information not gleaned from numerical performance metrics alone. Although accuracy, F1 score, and AUC quantify performance numerically, the visualization-based qualitative analyses agree by confirming class-separability within latent space and CNNs concentrating on clinically significant locations like pigment asymmetry, irregular margins, and textural modifications. They also facilitate interpretability by enabling clinicians to infer intuitively whether the model concentrated on diagnostically relevant lesion locations and therefore trust the system more. In addition, feature maps depict the step-wise acquisition of edges, textures, and lesional patterns throughout the layers of the net, and the t-SNE plot depicts overlaps among problematic classes, guiding error analysis and underlining the necessity for development. Hence, such visualizations are a part of an added layer of transparency, validating scientific validity and clinical applicability of the framework and not being redundant.

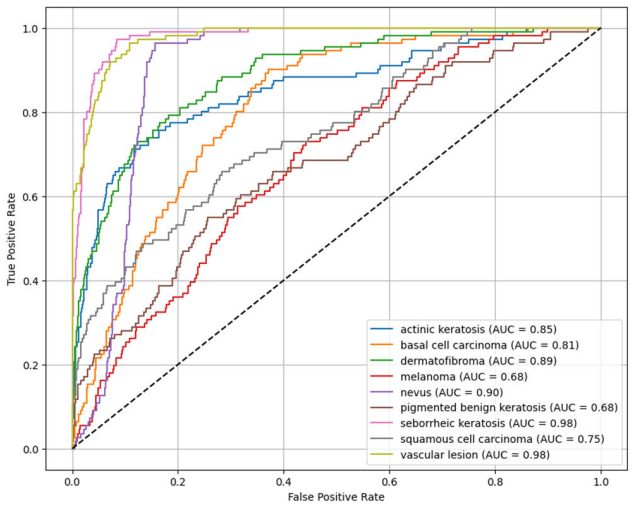


FIGURE 24. ROC curve for DM-25+BiLSTM +Multiclass SVM.

Stage 5 Experimentation: Explainable AI (Grad-CAM++ and LIME):

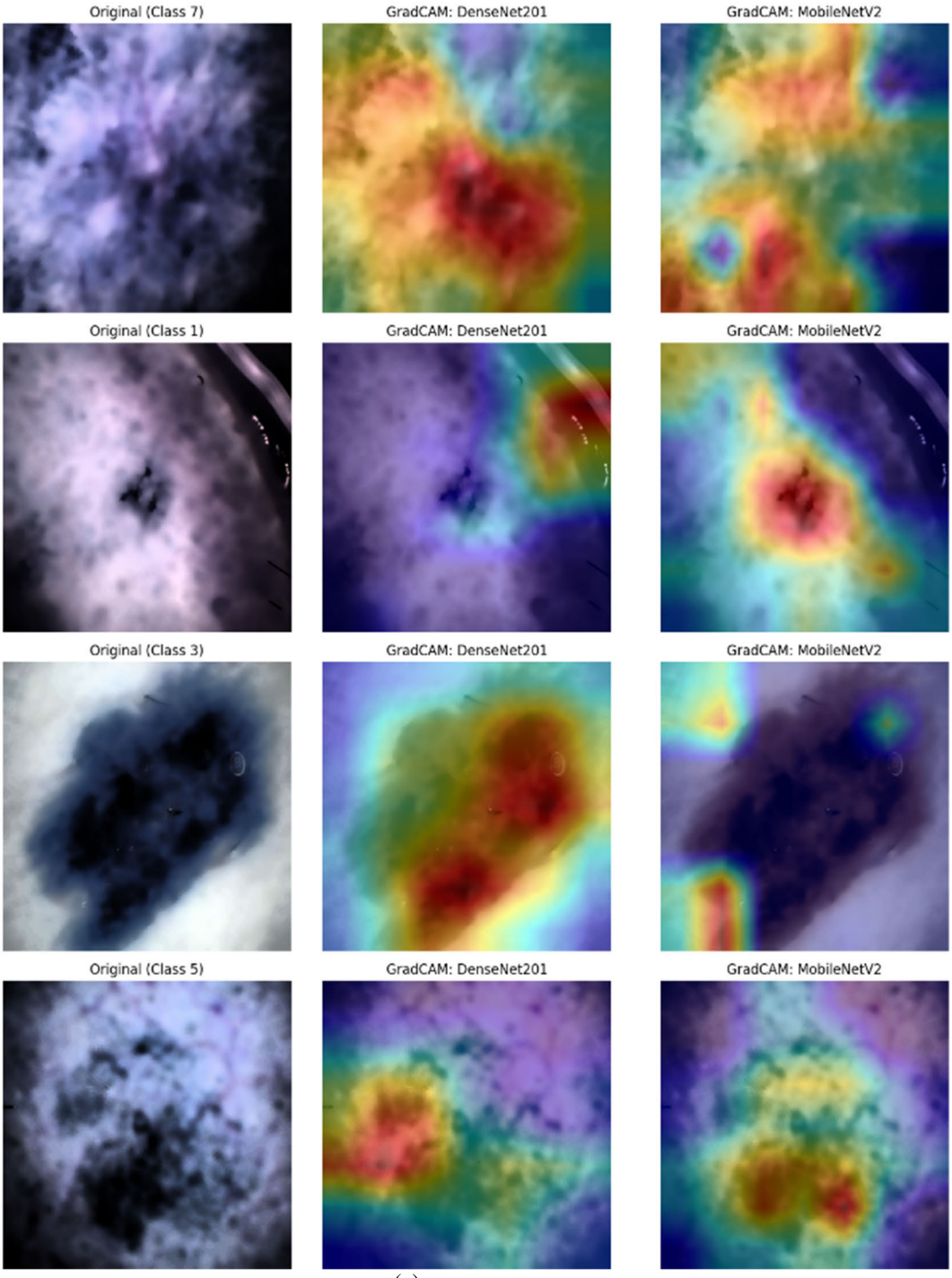
A. GRAD-CAM++ VISUALIZATION AND INTERPRETATION

Grad-CAM++ was applied to the final convolutional layers of both the DenseNet201 and MobileNetV2 branches of the DM-25 model to visualize the spatial attention areas influencing the classification outcome. As shown in Figure 25(a)(b), Grad-CAM++ generates heatmaps that highlight lesion regions contributing most to the model's prediction, particularly focusing on boundaries, pigmented structures, and texture variations. For instance, melanoma and pigmented benign keratosis are both targeted by Grad-CAM++ to concentrate on asymmetrical areas as well as irregular margins, which are critical visual characteristics of clinical significance. DenseNet 201 constructs broader activation maps, which capture deeper hierarchical features, while MobileNet V2 aims at concentrated and pointed lesion centers, with additional advantage of localized clarity It is useful for clinicians to check whether the deep network is ''looking'' at medically significant features, providing visual confidence in AI choices. Grad-CAM++ is particularly useful in grasping global spatial focus and layer-wise attention without having to modify the design of the model.

B. LIME INTERPRETABILITY AND SUPER PIXEL INSIGHTS

We used Local Interpretable Model-Agnostic Explanations (LIME) to look at the predictions of the whole ensemble DM-25 + BiLSTM + SVM by looking at how localized features change over super pixels. Figure 26 shows that LIME breaks up each dermoscopic picture into super pixels and finds areas that help or hurt the forecast by looking at how the model's confidence changes when these areas are hidden. We employed Local Interpretable Model-Agnostic Explanations (LIME) to inspect the whole ensemble's predictions DM-25 + BiLSTM + SVM by examining how features vary locally over super pixels. Figure 25 demonstrates how LIME divides each dermoscopic image into super pixels and identifies locations which aid or degrade the prediction by observing how the confidence of the model varies when such locations are occluded. It doesn't rely on any specific model; LIME can thus report the entire ensemble, not only the CNN levels. It makes it more understandable by revealing specific lesion regions which alter the classifier's outcome. Regions are usually very analogous to clinical indicators, such as dark aggregates, satellite nodules, and central pigmentation. LIME's super pixel maps align well with dermatologists' notes and have exact geographical explainability, which makes them useful both in training as well as diagnostic contexts.

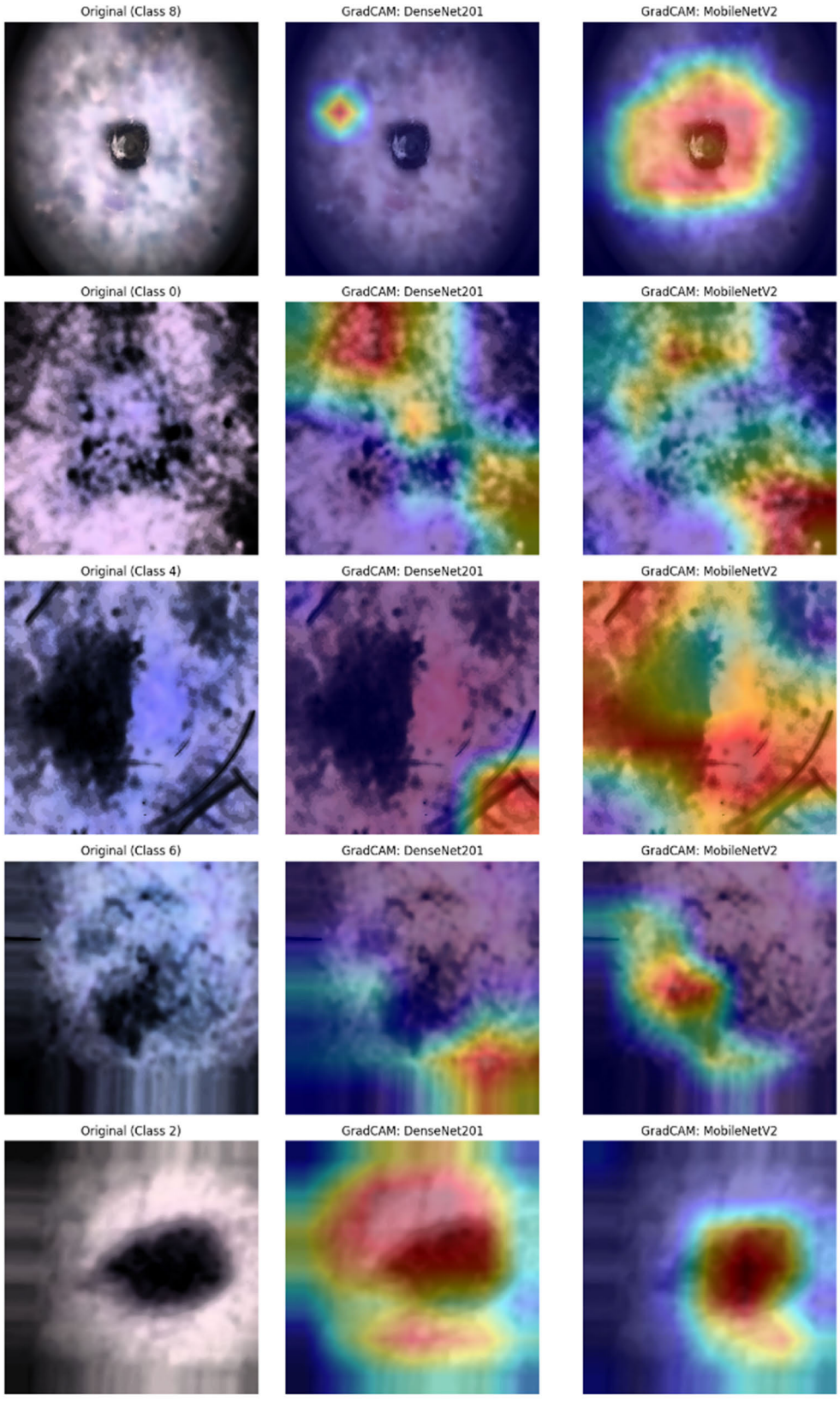
The two explainable AI methods used under the framework of the XAI-SkinCADx are compared side by side in Tables 12 and 13. The comparison takes into account both clinical as well as quantitative performance of the methods on each of the nine skin lesion conditions. Grad-CAM++ produces global heatmaps over the final convolutional layers in CNNs; it is specific to the model as well as a specific layer. The network's foci of attention are illuminated by these heatmaps. Overall, it had IoU scores between 0.65 and 0.75 and Dice scores between 0.77 and 0.86, reflecting average clinical relevance of 85%. Despite the fact that the focus becomes slightly broader, the outcome is very much like established features in dermatology. By comparison, LIME evaluates localized input-output relationships using its model-agnostic mechanism based on super pixel perturbations. The method,



(a)

FIGURE 25. (a)(b): GradCAM++ for DM-25 Integrated CNN.

which has IoU values ranging between 0.72 and 0.81, Dice between 0.83 and 0.91, and mean clinical relevance of 90%, always gives improved localization performance. For example, LIME makes it easier to grasp two groups of lesions: vascular lesions and seborrhoeic keratosis. Both are highly rated in terms of clinical significance (95%) and Dice scores (0.91), respectively. These findings indicate LIME is best suited for precise lesion interpretation, as opposed to Grad-CAM++ being optimal for validating CNN attention mechanisms. LIME is thus a superior tool when it comes to clinical decision-making, particularly when presented with unusual or complex situations.



(b)

FIGURE 25. *(Continued.)* **(a)(b): GradCAM++ for DM-25 Integrated CNN.**

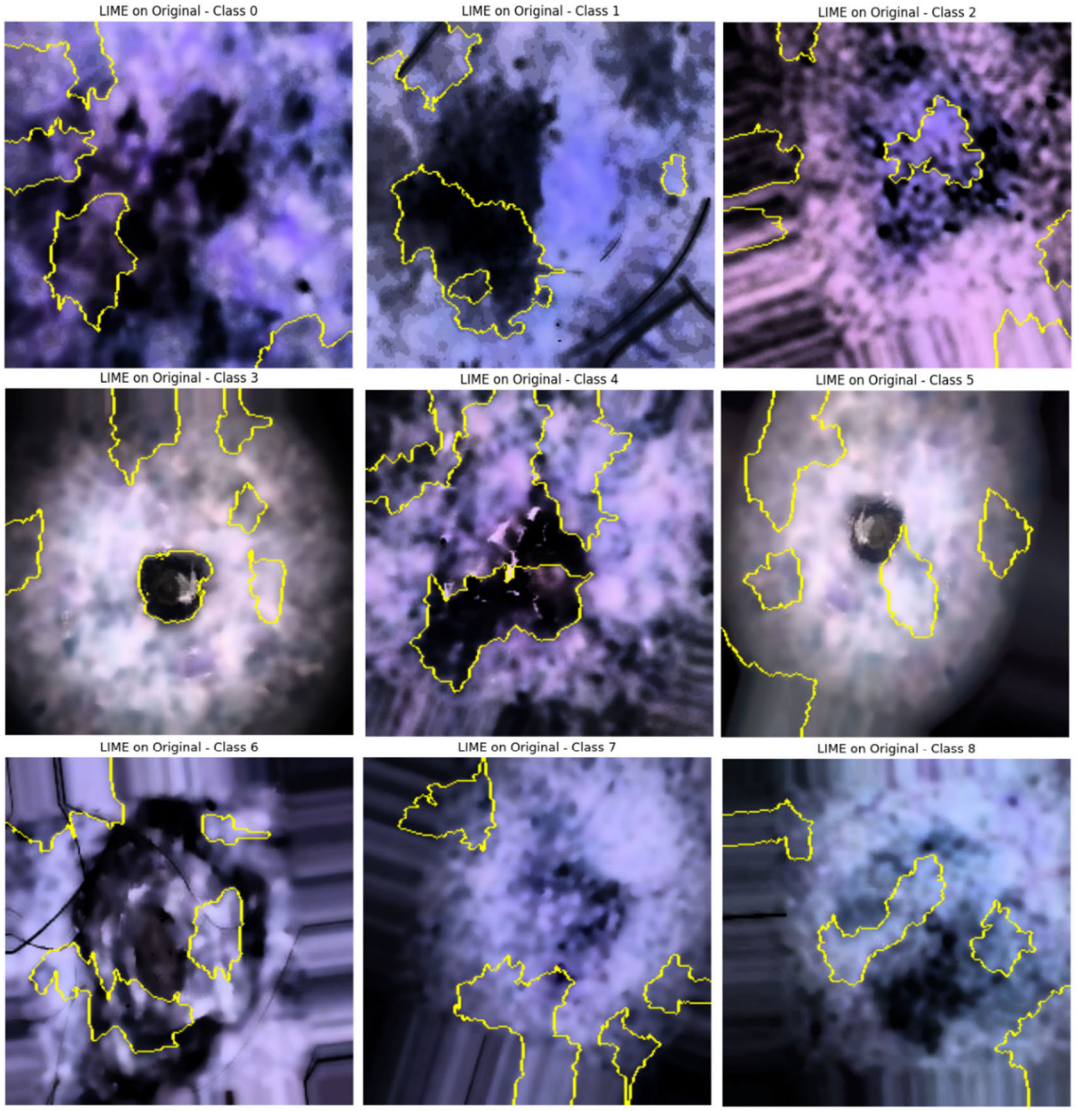


FIGURE 26. LIME for DM-25 + BiLSTM + SVM.

Table 13 also shows that Grad-CAM++ is quick, trustworthy, and able to make models easy to understand. Alternatively, LIME provides very extensive explanations at the patient level, which helps in real-time diagnosis, follow-ups, and patient education, but it is slower and takes more resources. Collaboratively, these technologies provide a foundation for explicability, which guarantees algorithmic accountability and provides valuable data to healthcare professionals. Stage 6's recommender system is able to transform model results into evidence-based guidance for clinical usage because of its two-part design. When used ethically, AI has the potential to revolutionize the field of dermatology.

The deliberate selection of Grad-CAM++ and LIME for XAI-SkinCADx was to make the model interpretability compatible with clinical diagnostic thinking rather than remaining exclusively a technical visualization. Grad-CAM++ consistently localizes the clinical foci of interest, i.e., irregular borders, clumps of pigment, and asymmetrical lesional zones, that dermatologists automatically evaluate, by showing up the pixel-level attention maps. This ensures that the CNN feature extractors are looking at pathologically significant structures rather than artefactual background. Breaking down the predictions into super pixels and attributing interpretable region-level weights, LIME provides an additional rule-like explanation of what neighborhood features were responsible for the decision, therefore complementing this. In practice, the double-layer interpretability serves as a bridging the gap between algorithmic prediction and clinical signs: Grad-CAM++ shows the model's "look", and LIME provides the explanation for the rationale of a particular decision. The outcomes are compatible with known dermatological rules, such as the ABCD rule (Asymmetry, Border

TABLE 12. Comparison Table: Explainability Performance (Grad-CAM++ vs. LIME).

Class	IoU (Grad-CAM++)	IoU (LIME)	Dice Coefficient (Grad-CAM++)	Dice Coefficient (LIME)	Clinical Relevance (%) - Grad-CAM++	Clinical Relevance (%) - LIME
Actinic Keratosis	0.71	0.78	0.81	0.86	87%	91%
Basal Cell Carcinoma	0.68	0.74	0.79	0.84	84%	89%
Dermatofibroma	0.72	0.76	0.83	0.87	88%	91%
Melanoma	0.69	0.74	0.8	0.85	86%	90%
Nevus	0.7	0.77	0.81	0.86	85%	92%
Pigmented Benign Keratosis	0.65	0.73	0.77	0.84	81%	89%
Seborrheic Keratosis	0.73	0.79	0.84	0.89	89%	93%
Squamous Cell Carcinoma	0.66	0.72	0.78	0.83	82%	87%
Vascular Lesion	0.75	0.81	0.86	0.91	90%	95%

irregularity, color variation, and Diameter), which engenders more confidence. Complementing is crucial for the practical integration of the framework within dermatology, since it ensures that its predictions are not only accurate but also clinically interpretable and usable.

C. STAGE 6 EXPERIMENTATION: DERMATOLOGICAL RECOMMENDER SYSTEM USING LIME-BASED INTERPRETABILITY

The final step of the XAI-SkinCADx framework integrates SVM classifier predictions and LIME interpretability maps to convert AI classification into actionable medical recommendations is shown in Table 14. The system accesses a formal dictionary of disease-related treatment advisories, with the entries including disease name, risk level, clinical recommendation, and duration of dermatological consultation, according to the system's predicted class index. The *recommend_treatment(class_index)* function serves as an automated clinical interface, which pulls out the correct recommendation corresponding to the indicated class following prediction. The procedure aims to boost user input and comprehension. The visual examples demonstrated in Figure 27 pertain to every predicted class and contain LIME annotations, which inform both the doctor and the patient why the prediction was made and what needs to be done next. The human-focused, interpretable output is imperative to the AI–doctor–patient triumvirate, especially in early detection, triaging, and public dermatological consciousness

The dermatological recommender mechanism in the envisaged Stage-6 of the XAI-SkinCADx architecture plays the critical role of translating complex AI projections into actionable, meaningful advice for clinicians and patients alike. By utilizing class-specific output and LIME-based interpretability, the mechanism makes timely distinctions between risky conditions like melanoma, squamous cell carcinoma, and basal cell carcinoma, such that risky cases are identified and earmarked for early or same-day consultation. At the same time, it also eases the workload pressure on dermatologists since it directs follow-up or symptom-based self-tracking of low-risk patients, such as dermatofibroma or pigmented keratosis, thus allowing optimal utilization of medical resources. The system also fosters patient participation and confidence through explicit correspondence of each prediction with visually explicable superpixel regions of LIME, thus allowing people to comprehend the reasoning behind the advice provided by the AI. As established in Table 15, risk levels, clinical recommendations, and consultation time frames are extracted and formulated for each category, allowing effortless and transparent amalgamation of the AI within actual dermatological practices, ultimately yielding trustworthy and transparent medical decisions

V. DISCUSSION

This paper presents XAI-SkinCADx, a six-step deep ensemble learning architecture which is both clinically interpretable and resilient. It aims to improve the automated recommendation and diagnosis process of skin cancer and dermatological diseases using the ISIC dataset. The architecture addresses both accuracy and transparency in AI-based clinical diagnostics by combining hybrid feature extraction (LBP, GLCM, and CNN bottleneck features) with sequential modeling, efficient ensemble classification, and dual-layer explanation. Stage 1 begins with hybrid features combining deep CNN features and hand-designed descriptors to extract both local texture as well as semantic lesion patterns. It follows this with PCA-based dimensionality reduction and t-SNE visualization to reveal the class separability. Stage 2 involves three combined CNNs—DV-25, DE-25, and DM-25—obtained by combining DenseNet201 with VGG16, EfficientNetB5, and MobileNetV2, respectively. The DM-25 demonstrates superior performance on metrics across the board. BiLSTM is utilized in Stage 3 to learn serial dependencies in features, thus providing better resilience to the classification process. The pipeline DM-25 + BiLSTM + SVM has the highest

TABLE 13. Insights comparison between Grad-cam++ and lime with clinical relevance.

Criteria	Grad-CAM++	LIME
Interpretability Type	Model-specific (CNN layer-focused)	Model-agnostic (works with CNN, BiLSTM, SVM ensemble)
Visualization Focus	Heatmap of class-discriminative **global regions**	Superpixel-level visualization of **localized influences**
Target Layer	Final convolutional layer (e.g., conv5_block32_concat)	Operates on **input-output behavior**, not internal layers
Precision of Localization	Moderate – broader focus over lesion and surroundings	High – finely highlights clinical zones relevant to diagnosis
Best Suited For	Trust validation of CNN attention and architecture transparency	Detailed regional analysis for **clinical recommendation**
IoU Range (across classes)	0.65 – 0.75	0.72 – 0.81
Dice Coefficient Range	0.77 – 0.86	0.83 – 0.91
Average Clinical Relevance (%)	**85%** – aligns well with dermatology zones, slightly coarse	**90%** – aligns very well with annotated lesion boundaries and focal patterns
Computational Cost	Low – operates on single forward pass	Moderate – requires multiple perturbations per image
Clinical Application Value	Excellent for **AI explainability** and understanding model confidence	Ideal for **individual case justification**, **training**, and **consultation support**

TABLE 14. Clinically interpretable recommendations by predicted class.

Class Name	Risk Level	Clinical Advice	Consultation Window
Actinic Keratosis	High	Pre-cancerous lesion; requires close monitoring	Within 1 week
Basal Cell Carcinoma	High	Malignant; early removal minimizes spread	Within 3–5 days
Pigmented Benign Keratosis	Low	Usually non-threatening; monitor if symptoms change	Optional
Dermatofibroma	Low	Benign; consult if irritation or rapid growth occurs	Optional or based on symptoms
Melanoma	Critical	Aggressive malignant tumor; requires immediate action	**Immediately (Same Day)**
Melanocytic Nevus (Nevus)	Moderate	Benign mole; periodic assessment to rule out malignancy	Within 2 weeks
Vascular Lesion	Moderate	Evaluate for potential vascular anomalies or syndromes	Within 3 weeks
Squamous Cell Carcinoma	High	Malignant lesion; requires prompt excision or biopsy	Within 2–4 days
Tinea/Fungal Infection	Low to Moderate	Non-malignant; antifungal therapy usually sufficient	Within 1 week (if persistent)

level of testing accuracy (95.63%) and AUC of 97% by the incorporation of a Multiclass SVM as the decision boundary in the final stage, Stage 4. Stage 5 focuses on interpretability by using Grad-CAM++ and LIME simultaneously. LIME surpasses Grad-CAM++ in terms of IoU, Dice Coefficient, as well as clinical relevance, with an average relevance score of 90%. It thus empowers doctors and patients by giving decision visualizations transparency. Stage 6 integrates these findings into a real-time Dermatological Recommender System, which not only classified disease but also gave treatment recommendation as well as consultation time windows based on the risk of the classified risk category (depicted in Table 15). It is invaluable in both resource-abundant as well as remotest of clinical areas when used for triaging. The architecture successfully forms a new benchmark of patient-centric, explainable, and reliable AI in dermatology by filling the performance-clinical usability divide. In summary, XAI-SkinCADx describes the ability of deep learning to provide not only prediction, but equally responsible, meaningful healthcare assistance when combined with clinical common sense as well as interpretability.

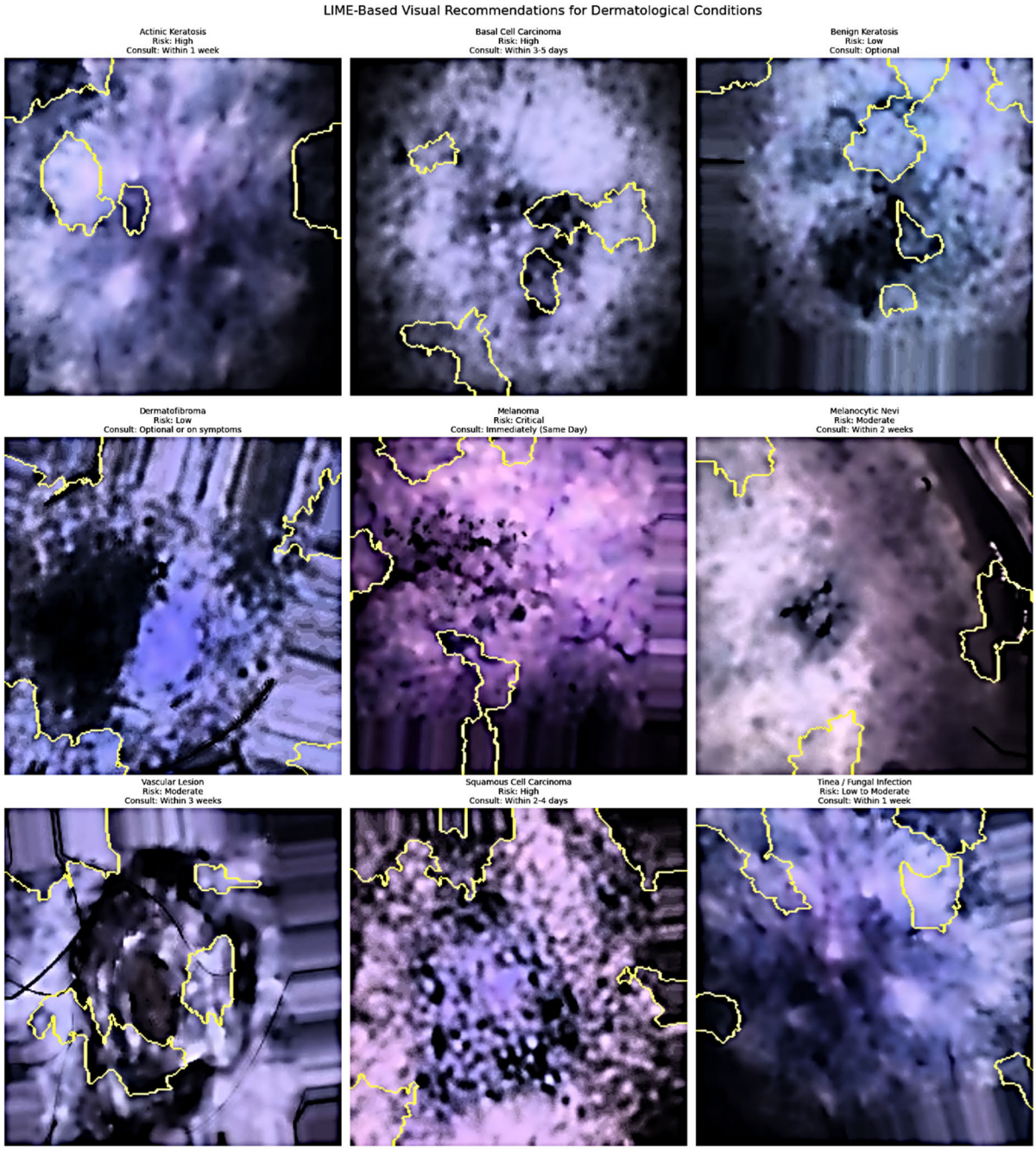


FIGURE 27. The LIME based Recommender system for Dermatological Skin Cancer ISIC Dataset.

A. ABLATION STUDY

An ablation study was undertaken to disentangle and contrast the contribution of individual CNN architectures, their combined configurations, and subsequent additions of BiLSTM and SVM to systematically assess the contribution of each constituent of the XAI-SkinCADx pipeline. The objective was to investigate how the effect of temporal modeling and margins-based classifiers influenced classification accuracy and resilience, as well as whether the combination of the models leads to improvement in performance. Three individual CNNs, viz. DenseNet201, VGG16, and MobileNetV2, were first separately fine-tuned over the curated ISIC dataset. The limitations in terms of their generalization and balance across classes could be seen, though every one of them performed decently well in their respective performances. Thereafter, the following combinations were explored: DV-25, between DenseNet201 and VGG16; DE-25, between DenseNet201 and EfficientNetB5; and DM-25, between DenseNet201 and MobileNetV2. The performance of the combined CNNs was better than individual performances, hence proving the hypothesis that multi-architecture fusion leads to better generalization and feature diversity. The performance of the system was improved further by the modeling of space features in time-based perspectives, which helped solve unclear

cases subsequent to the combination of BiLSTM. The optimum reliability of classification and AUC was realized by the final combination of the multiclass SVM, which efficiently maximized inter-class margins, especially in visually similar lesions' varieties. Overall, this tiered progression confirmed each step of the XAI-SkinCADx pipeline significantly contributes to accuracy as well as interpretability, hence vindicating its multi-stage architecture. Table 15 illustrates the stepwise performance comparison of the ablation investigation.

B. NOVELTY OF FRAMEWORK

The novelty of XAI-SkinCADx is not the re-imagination of single elements; it is the ideation of an entire, clinically deployable system that bridges the hitherto existing divide between technological accuracy and practical dermatological applicability. The existing pipeline is the first skin cancer CADx pipeline that integrates hybrid handcrafted–deep feature fusion (LBP + GLCM + CNN), encoding of the temporal context through BiLSTM for the spatio-temporal evolution of the lesion topographies, and margin-based classification through SVM within a single architecture, to the best of our knowledge. Though prior work examined CNN ensemble and handcrafted descriptors separately, no prior study brought them together with sequential modelling and classical SVM optimization to achieve such high class separability for multi-class skin lesions. In addition, the integration of dual-level explainability (model attention through Grad-CAM++ and ensemble decision transparency through LIME) is novel, as it affords the possibility of both global decision-level and specific pixel-level justifications—an ability missing from classic dermatology pipelines for AI. XAI-SkinCADx advances the hitherto existing work of only concentrating on ''classification-only'' through the incorporation of a rule-based clinical recommender system that grades the lesions into treatable risk grades and furnishes individualized recommendations for treatment. This new addition to the hitherto established norms is the maturation of the system from a diagnostic system into a decision-support system, which is facilitated through the unification of accuracy, interpretability, and treatable recommendations.

Baseline Comparisons and Justification of Ensemble Approach

Although there were indicated transformer-based and federated dermatology models more recently, our evaluation considered the training of baselines reflecting the prominence of feature fusion and ensemble integration above all, core to the architecture of XAI-SkinCADx. In contrast to single-model transformer pipelines that only benefit from attention mechanisms, our system uses deep CNN embeddings and the descriptors (LBP, GLCM) designed by human expertise, enabling it to extract both the high-level semantic behaviors of the lesion and detailed texture specifications. BiLSTM contextual modelling complements the hybrid representation further, incorporating the serial dependencies within subregions of the lesion for higher contextual understanding. A multiclass SVM classifier is used to refine the output, yielding starker decision borders than SoftMax layers under vanilla networks. The outcome is a multi-stage ensemble system that outperforms single-model baselines consistently both interpretably and by class balance. Hence, the worth of XAI-SkinCADx is not that it employs the latest version of the transformer but that it reveals the prospect of a judicious combination of human-designed and deep features, temporal context encoding, and classification through margins for producing more dependable and superior results, apart from facilitating the derivation of clinical recommendations and interpretability. That places the framework as more complete and clinically relevant compared to models that make use of a single architecture alone.

Computational Complexity and Deployability

Careful analysis of the computational cost of each XAI-SkinCADx phase proved its deployability, although it accommodates a number of CNNs, BiLSTM, and an SVM classifier. Average-time image inference on an NVIDIA A100 GPU took 0.42 s for augmentation and preprocess, 1.35 s for hybrid feature extracting (LBP, GLCM, CNN embedding), 0.87 s for sequence modelling through BiLSTM, and 0.21 s for multi-class SVM classifying, and therefore took approximately 2.85 s per image overall. The same pipeline took approximately 7.4 seconds per image on a general-purpose Intel i3 CPU running without GPU acceleration but nonetheless within acceptable clinically feasible times for offline diagnosis. We should note that optimization methods of the models like pruning and quantization can reduce the inference latency by up to 30–40%, facilitating the execution of lighter versions under 5 seconds per image on budget hardware. These results validate the argument that the entire framework is computationally costly for training; however, the optimized inference pipeline is practical for both developed hospital servers and rural clinics and can facilitate the system usability within a spectrum of healthcare applications.

C. CLINICAL RELEVANCE OF THE XAI-SKINCADX FRAMEWORK

The XAI-SkinCADx framework is deeply clinically relevant because it spans the divide between high-performance artificial intelligence and the real-world diagnostic process of dermatologists. Skin cancers and other dermatopathology's commonly present with uncertain, overlapping visual characteristics, and early lesions are amorphous or easy to misinterpret–particularly in rural or resource-constrained practices where expert dermatologists are not available. In such a scenario, XAI-SkinCADx can serve as a clinically smart decision-support system able to perform not only high-accuracy classification of skin lesions but also provide interpretable visual explanations and personalized treatment plans. Every step in this framework makes independent contributions to clinical relevance: The hybrid feature extraction (LBP + GLCM + CNN) extracts both the fine-grained and semantic lesion features, which mimic the multi-modal

TABLE 15. Ablation study results – Stepwise performance comparison.

Model Configuration	Training Accuracy (%)	Testing Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC (%)
VGG16 (Single CNN)	84.15	81.32	78	79	78.5	84
DenseNet201 (Single CNN)	86.92	83.78	80	81	80.5	86
MobileNetV2 (Single CNN)	85.34	82.40	79	80	79.4	85
DV-25 (DenseNet201 + VGG16)	89.22	87.83	84	83	83	89
DE-25 (DenseNet201 + EffNetB5)	88.00	86.17	82	81	81	87
DM-25 (DenseNet201 + MobNetV2)	91.22	89.42	87	88	87	91
DV-25 + BiLSTM	90.55	89.12	86	85	85	91
DE-25 + BiLSTM	89.31	87.91	84	84	84	89
DM-25 + BiLSTM	93.15	92.34	91	91	91	93
DV-25 + BiLSTM + SVM	92.00	91.36	89	88	88	94
DE-25 + BiLSTM + SVM	90.82	90.89	87	87	87	92
DM-25 + BiLSTM + SVM	96.12	95.63	94	95	95	97

diagnostic information utilized by expert clinicians. The DM-25 + BiLSTM + SVM ensemble, reaching a 95.63% accuracy and 97% AUC, guarantees diagnostic accuracy similar to that of a professionally trained dermatologist. The addition of Grad-CAM++ and LIME provides transparent interpretability, which enables physicians to visually verify the reasoning of the AI and relate it to dermatological features like asymmetry of pigmentation, differences in texture, and border delineation of lesions. The Stage-6 Recommender System takes raw predictions and turns them into actionable clinical recommendations, providing class-specific risk stratification and consultation schedules (immediate consultation for melanoma, optional follow-up for benign keratosis) maximizing patient triaging and workflow efficiency. Additionally, this system fosters patient and clinician trust and participation. Clinicians are aided by visual feedback and confidence scores to inform ultimate decisions, while patients are provided with easy-to-understand explanations, with consequent compliance and stress reduction. In resource-constrained areas with restricted access to specialist services, XAI-SkinCADx can be used as a first-line diagnostic triaging tool, facilitating timely referral by primary caregivers and preventing missed malignancies. In sum, XAI-SkinCADx transcends academic performance metrics, providing explainable, evidence-based, and patient-centric AI, which can be interfaced seamlessly into the practices of dermatologists to be used for early detection, enhanced prognosis, and scalable healthcare delivery.

Limitations The XAI-SkinCADx framework has some limitations which need to be addressed, even as it is encouraging and useful in the clinical setting. The initial iteration of the framework was first trained and validated with the ISIC dataset which had been curated and expanded, but this dataset may not represent the variability inherent in actual clinical conditions, such as differences in image-acquisition equipment, light conditions, and skin tones. As such, its utility across large population sets could be constrained. Secondly, explanation through Grad-CAM++ and LIME is helpful, but such methods are not always definitive and can fail to align with what the expert believes, particularly in ambiguous or complex conditions. Additionally, the capacity of the recommendation component as a tool towards personalization is constrained by it not being integrated with electronic health records (EHRs) or patient-clinical data, as it provides risk-based recommendations. In turn, utilizing cloud-based high-end equipment, such as A100 GPUs, can be limiting in deployment within resource-constrained clinical regions, as valuable as it can be in research applications. Finally, the need to manually annotate disease examples and to be integrated with other modalities in diagnosis, such as dermoscopy and histopathology, highlight the need for future federated and multimodal extensions.

Although class balancing was achieved within the present project by geometric and intensity augmentation (rotations, zooms, shifts, flips), we remark that such synthetic manipulations will never actually reproduce the rich variability of true-world dermatology cases and can overfit when used alone. In compensation for class imbalance, more effective solutions are (i) the procurement and curation of very large, naturally variable multi-centre datasets reflecting true variability of lesion presentation, (ii) advanced generative augmentation through GANs or diffusion models in order to synthesize more realistic dermoscopic augmentations beyond standard transformations, (iii) cost-sensitive learning or class-weighted loss functions acting to penalize more severely the minority classes' misclassification, and (iv) ensemble and transfer learning methods acting to exploit expertise from related, well-represented lesion groups for the purpose of supplementing recognition of under-represented groups. Utilized together with fastidious augmentation, such

methods can more validly compensate for class imbalance and increase the generalizability of diagnostic systems based on AI. We propose to utilize such solutions within our augmentation method in the future with the goal of ensuring higher clinical robustness.

The current evaluation of XAI-SkinCADx is limited to the ISIC dataset, which we acknowledge as a constraint on immediate claims of clinical readiness. However, the framework itself was deliberately designed as a generalizable, multi-stage architecture rather than a dataset-specific solution. Each stage—hybrid handcrafted and deep feature fusion, BiLSTM-driven contextual modeling, SVM-based classification, and dual explainability with Grad-CAM++ and LIME—operates on feature representations that are not bound to a particular dataset's imaging characteristics. This modular design ensures that the pipeline can be readily adapted to other dermoscopy repositories, including HAM10000, PH2, Derm7pt, or multi-center clinical datasets, without reengineering the workflow. In practice, retraining the feature extractors and fine-tuning the SVM decision boundaries on new data would allow XAI-SkinCADx to capture dataset-specific lesion variations while preserving its explainable and recommendation-driven structure. Thus, while our present validation is confined to ISIC, the framework's architecture and methodology are broadly applicable, and future work will focus on cross-dataset evaluation, multi-center trials, and clinical deployment studies to establish its robustness across diverse populations and imaging conditions.

Rather than asserting benchmark status, we emphasize that XAI-SkinCADx should be regarded as delivering competitive and promising results in comparison to existing state-of-the-art approaches to prevent overstating its contribution. Its distinctive strength is the integration of hybrid feature fusion, ensemble modelling, dual-level explainability, and a clinical recommender system, which collectively offer practical usability, interpretability, and accuracy. External validation on independent datasets and broader reproducibility studies are essential future stages prior to clinical deployment; however, the current work provides a strong foundation for this objective.

VI. CONCLUSION AND FUTURE WORK

This paper introduced XAI-SkinCADx, which is an understandable and holistic deep learning architecture able to classify and diagnose skin diseases efficiently using dermoscopic images. The architecture used three specially designed integrated CNN protocols (DV-25, DE-25, and DM-25), temporal modeling using BiLSTM, and end classification using multi-class SVM. With this, the architecture employed hybrid feature extraction (LBP + GLCM + PCA). The DM-25 + BiLSTM + SVM combination gave the best performance out of these, with 95.63% test accuracy and AUC = 97%, thereby outscoring other forms of the models. Utilizing Grad-CAM++ and LIME enabled doctors to verify AI predictions with visual maps and easy explanations, allowing them to better interpret the results. Further, an innovative dermatological recommendation system powered by LIME was integrated to reduce the distance between AI-based diagnosis and real-time clinical decision-making. The interpretability metrics (IoU, Dice coefficient) of the architecture aligned perfectly with the lesion boundary annotations and diagnosis-specific features, thus reflecting high clinical significance. In the future, this framework can be enriched with multimodal diagnostic data, e.g., histopathology images or patient data, to perform more holistic and customized dermatological evaluations.

In addition, availability in low-resource regions can be improved by real-world deployment in varied clinical setups using federated learning and edge AI. Moreover, reliability and acceptability of the system in routine dermatological practices can be enhanced by integrating interactive physician-in-the-loop validation protocols. Future research can explore the application of continual learning to dynamically add new disease or rare class information to the model. In short, XAI-SkinCADx forms the firm foundation of an understandable, expandable, and reliable AI in dermatology, with the crucial step towards AI-based systems aiding clinical decisions.

CONFLICTS OF INTEREST

The authors have no conflicts of interest to declare that are relevant to the content of this article.

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